Action-Conditioned Predictive Consistency as a Diagnostic for Gaussian-Noise Robustness in JEPA World Models

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Abstract

Latent predictive world models avoid pixel reconstruction, but pixel-free prediction is not a robustness definition for closed-loop control. We study this gap with action-conditioned predictive consistency (ACPC), a no-retraining diagnostic for JEPA-style control checkpoints. ACPC asks whether clean and corrupted views of the same state produce consistent future predictions under the same action rollout, while task-grounded near-boundary pairs test that action-relevant distinctions remain separated. We calibrate the readouts through fixed-candidate cost drift, margin-failure, and local sensitivity analyses, without claiming closed-loop guarantees. Across three independent training seeds and four control tasks, no-noise LeWM checkpoints are fragile under observation-only Gaussian noise, while input-side noise training yields broad matched-Gaussian robustness plateaus. Diagnostic scores fixed on one development seed identify held-out candidate regions enriched for plateau members; they do not rank checkpoints inside the plateau. Margin and planner-trace audits support the selective interpretation: recovered checkpoints contract same-state noisy rollouts while preserving task-grounded near-boundary separations and reducing clean/noisy optimizer sensitivity. Blur/resize checks remain bounded scope evidence: diagnostics track clear TwoRoom/Reacher gains and stay weak or mixed when PushT/Cube show no clear gain. The contribution is a controlled no-retraining diagnostic for JEPA robustness plateaus, not a universal transfer or training-objective claim.

Keywords: visual world models; JEPA; action-conditioned predictive consistency; visual robustness; predictive-dynamics diagnostics.

Code and data release: `https://github.com/Anguo-star/lewm-acpc-diagnostics`

1 Introduction

1.1 Latent prediction is not a robustness definition

JEPAs predict future representations instead of reconstructing pixels, which can reduce pressure to encode high-magnitude irrelevant visual detail [2, 4, 39]. For control, however, "irrelevant" is action- and transition-dependent. A representation may ignore a background texture but must preserve a contact pose or doorway state if that changes the next useful action. Thus encoder-level invariance is an incomplete target: planning uses the model's *predictions*, so robustness must be stated after action-conditioned prediction.

We use this failure as a controlled diagnostic probe. Across three independent LeWM [26] training seeds, the no-noise baseline remains strong on clean observations but loses 74.56 ± 5.55 points on PushT and 41.00 ± 1.44 points on Reacher under observation-only Gaussian noise at std 0.08 with the goal unperturbed. Observation+goal noise is reported as a harder auxiliary stress condition. Test-time visual fragility is not new [21, 43, 16, 28]; our focus is what it reveals about JEPA latent world-model control.

1.2 Operationalizing action-conditioned predictive consistency

We operationalize a downstream-control view of visual robustness for JEPA world models through *action-conditioned predictive dynamics*. Let $z = E(h)$ and $\tilde{z} = E(\tilde{h})$ be the latents an encoder E produces from an unperturbed history h and a perturbed history $\tilde{h}$ of the same underlying state, and let $F^k(\cdot, \mathbf{a}_{t:t+k-1})$ be the action-conditioned latent predictor rolled out for k steps under an action sequence $\mathbf{a}$. We *allow* $z \neq \tilde{z}$. The robustness target is placed *after* the predictor: in task-relevant coordinates,

$$F^k\big(z, \mathbf{a}_{t:t+k-1}\big) \;\approx\; F^k\big(\tilde{z}, \mathbf{a}_{t:t+k-1}\big), \qquad k = 1, \ldots, H, \tag{1}$$

so that the two branches induce the same next-state and rollout predictions under the same action intervention. The requirement is *selective*: it should hold only for nuisance perturbations of the same underlying state. State differences that change action-conditioned transitions, costs, or the optimal action must remain *discriminable*, or the planner loses the distinctions it needs. We make both halves precise in Section 3.

This diagnostic view changes what counts as a useful robustness measurement. Encoder geometry is a useful first-stage diagnostic because it shows whether visual perturbations enter and reshape the latent neighborhood, but encoder-level latent closeness, $\|z - \tilde{z}\| \to 0$, is not a complete control-robustness definition: large nuisance shifts can wash out after prediction, whereas small shifts can still be amplified downstream. Planning, cost, and action metrics remain useful as *downstream readouts* of predictive consistency; future objectives should regularize action-conditioned predictive dynamics while preserving action-relevant discriminability.

1.3 Existing noisy-evaluation results as a controlled probe

We use input-side noise augmentation as the controlled intervention. The $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$ sweep across four tasks creates a reproducible grid of paired baseline/recovered checkpoints for asking where closed-loop recovery appears in the encoder–predictor–planner chain. Because the sweep forms broad task-dependent plateaus under finite evaluation variance, we do not treat a single point-best $\sigma_{\max}$ or small within-plateau score differences as meaningful checkpoint ordering. Appendix N records a negative target-view ablation: perturbed-history $\to$ original-future denoising is not a sufficient closed-loop fix, so the retained empirical branch is full-sequence perturbed-target training.

1.4 Contributions

C1 — Control-facing diagnostic principle. We operationalize visual robustness for JEPA world-model control as a staged diagnostic: encoder geometry shows whether visual noise enters the latent state; ACPC asks whether the shift changes action-conditioned futures and candidate costs; a discriminability countercondition checks that action-, transition-, or cost-distinct states remain separated.

C2 — Calibration link. Under a Lipschitz cost readout, bounded rollout disagreement bounds fixed-candidate cost drift. For a once-sampled candidate pool, top-1 instability decomposes into ACPC-tail and clean-margin terms. A local Gaussian linearization then shows that the measured basin radius estimates action-conditioned encoder–predictor sensitivity $\|J_G J_E\|_F$, not encoder invariance alone. These results justify the readouts as diagnostics while leaving adaptive CEM resampling, repeated replanning, and closed-loop trajectories to empirical tests.

C3 — Fixed-checkpoint evidence chain. On fixed checkpoints, we report a three-training-seed evidence chain: matched-Gaussian recovery; a frozen diagnostic plateau-membership screen; selective-margin and planner-side checks; and bounded blur/resize scope checks. Larger grids, PLDM replication, ablations, and artifact paths remain in the appendix and release manifest.

2 Related Work

2.1 Latent prediction and JEPA world models

JEPA [23] predicts in latent space rather than reconstructing pixels; I-JEPA [2] and the V-JEPA family [4, 3, 27] extend this idea to images, video, and dense physical-world representations. LeWM [26], our baseline, stabilizes an end-to-end JEPA world model with SIGReg [8]. PLDM [33, 34], via `stable-worldmodel` [25], supplies the second method family in Appendix H. LeJEPA theory [19] studies latent-variable recovery and latent planning; our diagnostic asks whether learned visual world-model checkpoints give paired clean/noised views consistent action-conditioned predictions.

2.2 Invariance, augmentation, and visual robustness

Augmentation and joint-embedding losses impose invariances only when their training signal identifies the right nuisance factors [41, 36, 45]. Recent theory clarifies why latent prediction can help with nuisance or noisy features but also why it is not enough for control. Van Assel et al. [39] derive closed-form SSL solutions showing that joint embedding imposes weaker alignment requirements than reconstruction when irrelevant features have large magnitude, while Littwin et al. [24] analyze deep-linear self-distillation dynamics and a JEPA bias toward high-influence predictive features rather than merely high-variance features. These analyses are representation-level and do not address closed-loop action-conditioned rollout, candidate-cost stability, or the need to keep action-distinct states separable. ACPC is complementary: it asks whether same-state visual perturbations become equivalent after a fixed action intervention while action-, transition-, or cost-distinct cases remain discriminable. Seq-JEPA [12] addresses a related invariance–equivariance tension architecturally. DrQ/DrQ-v2 [21, 43], SODA/DMC-GB [16], DreamerV3 [14], TD-MPC2 [15], ViGMO [28], N-JEPA [29], variational VJEPA [17], and US-JEPA [30] provide visual-robustness and noisy-environment context. They are not trained as competing baselines in this study; they define important future comparisons for method or benchmark claims.

2.3 Control-relevant predictive representations

Bisimulation merges states with equivalent reward and transition behavior. DeepMDP [11], DBC [44], bisimulation-regularized MPC [32], and Bisim-JEPA [38] use this principle to learn control-relevant representations. Value-equivalent and value-aware models [13, 40] and VIBR [7] likewise emphasize downstream control consequences. Wang et al. [42] formalize action-conditioned video world modeling as a group action with identity, inverse, and composition-consistency metrics. ReOI [6] treats visual distractor robustness in visual MPC through observation interventions for world-model action-outcome prediction. These lines show that downstream predictive consequences are already central to control-representation work. Our diagnostic instantiates that view for paired clean/noised JEPA latent-world-model rollouts: same-state noised and unperturbed views should agree after the same action sequence, while action-distinct transitions remain separable.

Our use of ACPC is therefore narrower than an identifiability theorem, a learned bisimulation metric, or a robust MPC intervention. LeJEPA-style results motivate latent prediction and planning under latent recovery assumptions; bisimulation and value-aware models define task abstractions; group-action metrics test action faithfulness; ReOI modifies observation handling for robust visual MPC. This paper instead supplies a JEPA checkpoint diagnostic centered on additive Gaussian pixel-noise stress tests: predictive stability is tested on paired same-state clean/noised rollouts under the same action sequence, fixed-candidate cost stability is derived as a sufficient condition, and a separate discriminability guard checks that action-, transition-, or cost-distinct cases have not been collapsed.

2.4 Representation diagnostics and collapse analysis

We use standard representation diagnostics: effective rank [31, 18, 37], CKA [20], linear probes [1], and anti-collapse context from VICReg [5]. RankMe [10] motivates treating representation geometry as a diagnostic rather than as a release criterion. The individual metrics are standard; our contribution is the protocol that ties them to closed-loop Gaussian-noise evaluation and ACPC-style paired checks, not another standalone representation score.

3 Action-Conditioned Predictive Consistency

This section defines action-conditioned predictive consistency and its discriminability countercondition. We use ACPC to separate two requirements that encoder invariance alone conflates: same-state visual perturbations should agree after action-conditioned rollout, while action-, transition-, or cost-distinct states should remain separable. Section 3.2 then separates the main descriptive diagnostics from paired ACPC probes.

3.1 Setup and definitions

Let o_t be an unperturbed observation and $\tilde{o}_t = \tau(o_t)$ a visually perturbed view of the *same* underlying state (τ is additive Gaussian pixel noise here, but the definitions are perturbation-agnostic). Let h_t and $\tilde{h}_t$ be the corresponding observation–action histories, E_θ the encoder, and F_θ the action-conditioned latent predictor. Fix an action sequence $\mathbf{a}_t^H = a_{t:t+H-1}$ that is *identical* on both branches. For compact notation below, $\mathbf{a}_{0:k-1}$ denotes the length-k prefix of this sequence after re-indexing time at t, and we write

$$z_t = E_\theta(h_t), \quad \tilde{z}_t = E_\theta(\tilde{h}_t), \qquad \hat{z}_{t+k} = F_\theta^k(z_t, \mathbf{a}_{0:k-1}), \quad \hat{\tilde{z}}_{t+k} = F_\theta^k(\tilde{z}_t, \mathbf{a}_{0:k-1}). \tag{2}$$

Let Π denote a *task-relevant predictive readout* — the full latent, a transition delta, goal-distance or cost features, or a learned projection — and let d be a distance on its range.

Action-conditioned predictive consistency. For a same-state perturbation pair under the shared action sequence,

$$\mathrm{ACPC}_H(o_t, \tilde{o}_t, \mathbf{a}) = \sum_{k=1}^{H} \alpha_k\, d\big(\Pi(\hat{z}_{t+k}),\, \Pi(\hat{\tilde{z}}_{t+k})\big), \qquad \alpha_k \geq 0. \tag{3}$$

For the theoretical statements below, we use the induced horizon metric

$$d_H\big((u_1, \ldots, u_H), (v_1, \ldots, v_H)\big) = \sum_{k=1}^{H} \alpha_k d(u_k, v_k), \tag{4}$$

so that Equation (3) is exactly the clean/perturbed rollout-readout distance under the shared action sequence. In experiments we instantiate this with the L2 distance over rollout tokens used by R_F. In this diagnostic framing, robustness corresponds to small ACPC_H together with action-relevant discriminability, not to the encoder distance $d(z_t, \tilde{z}_t)$ alone. Encoder closeness is a useful first-stage risk signal, but it is neither necessary (a large but task-irrelevant encoder shift can wash out through F_θ and Π) nor sufficient (a small encoder shift can still change the predicted future enough to flip the planned action).

Action-relevant discriminability (countercondition). Consistency must be *selective.* Let $Y^H(s, \mathbf{a})$ denote a task-relevant future readout that is external to the learned predictor — simulator state, cost-to-go, inverse-dynamics label, contact/keyframe tag, or another diagnostic proxy available in the dataset. For two genuinely different states s_i, s_j (with latents z_i, z_j) that admit an action sequence under which this external future readout diverges,

$$\Delta^H_{\mathrm{dyn}}(s_i, s_j, \mathbf{a}) = d_Y\big(Y^H(s_i, \mathbf{a}),\, Y^H(s_j, \mathbf{a})\big) > m, \tag{5}$$

the representation and predictions should keep them separable,

$$d\big(\Psi(z_i),\, \Psi(z_j)\big) > m', \tag{6}$$

where Ψ is a discriminability readout, possibly action-conditioned (latent, predicted delta under $\mathbf{a}$, inverse-dynamics feature, cost feature, or rollout embedding), and $m, m' > 0$ are margins. Equations (3)–(6) state the target: lower predictive disagreement for same-state visual perturbations must preserve discriminability among state/action/transition-distinct pairs. A method that drove $\mathrm{ACPC}_H \to 0$ by collapsing the latent would violate (6), so the two conditions are inseparable.

3.2 Operational consistency diagnostics

The main paper uses one fixed ACPC instance. For the Gaussian-noise basin diagnostic, Π is the identity readout on the model inference/cost latent rollout, d is Euclidean L2 over rollout tokens, and $H = 8$. The reported quantities are R_E, the median same-state encoder-history spread normalized by the original-view nearest-neighbor L2 scale, and R_F, the median same-state rollout-latent spread normalized by the unperturbed transition L2 reference over the same horizon. R_F/R_E is a descriptive contraction ratio. This empirical R_F is a distributional, recorded-action localization proxy. It should not be read as an estimate of the uniform ϵ in the fixed-candidate stability corollary, which requires a bound over every candidate in a shared set $\mathcal{A}$. A compact shared-candidate table reports paired candidate-cost change (PCC), candidate-ranking agreement (CRA), and margin-conditioned action-flip rate (MAF) in Section 5.5; the full LeWM/PLDM readout artifacts are listed in `DATA_MANIFEST.md`.

Thus R_E and R_F answer different questions. R_E measures whether the visual perturbation changes the encoded neighborhood; R_F measures whether that change survives action-conditioned rollout in coordinates used for planning. Robust checkpoints in this Gaussian-noise study often reduce both; the diagnostic point is to interpret encoder contraction through its downstream predictive effect and the accompanying discriminability guard.

The main text reports three diagnostics aligned with the evidence chain: closed-loop success/drop for the behavioral effect, the ACPC rollout/cost family (R_F, PCC/CRA/MAF, margin-conditioned flips, and a small CEM trace audit) for action-conditioned predictive stability, and the task-grounded near-boundary proxy margin for selective discriminability. Auxiliary ADM/SPRR, latent-noise, artifact-key, and full representation diagnostics remain in the appendices.

3.3 ACPC as a fixed-candidate stability condition

ACPC is connected to planning through the candidate costs evaluated by model-predictive control. Let h be a clean history, $\tilde{h}$ the corresponding perturbed history, and let both branches evaluate the same candidate action sequence $\mathbf{a}$. For a goal or task descriptor g, define

$$C_h(\mathbf{a}, g) = J(\Pi(\hat{z}_{t+1}), \ldots, \Pi(\hat{z}_{t+H}), g)\,, \tag{7}$$

where J is the planner's cost readout on the rollout-readout sequence. $C_{\tilde{h}}(\mathbf{a}, g)$ is defined analogously from the perturbed branch. The following statements are sufficient conditions for stability on a fixed candidate set; they do not prove stability of the full CEM sampling distribution or of the closed-loop trajectory over repeated replanning.

Proposition 1 (Encoder shift as one route to rollout disagreement)**.** *Fix an action sequence* $\mathbf{a}$ *and write*

$$G_{\mathbf{a}}(z) = \Pi(F_\theta^{1:H}(z, \mathbf{a}))$$

for the rollout-readout map. If $G_{\mathbf{a}}$ *is locally* L_G*-Lipschitz between* $E_\theta(h)$ *and* $E_\theta(\tilde{h})$ *under encoder metric* d_Z *and rollout-readout metric* d_H*, then*

$$d_H\big(G_{\mathbf{a}}(E_\theta(h)), G_{\mathbf{a}}(E_\theta(\tilde{h}))\big) \leq L_G\, d_Z\big(E_\theta(h), E_\theta(\tilde{h})\big)\,.$$

Proof. This is the local Lipschitz condition applied to the two encoded histories. □

This bound makes encoder geometry a first-stage risk signal: small encoder shifts imply small rollout disagreement only under local insensitivity, large shifts can be contracted by $G_{\mathbf{a}}$, and small shifts can be amplified by the predictor or cost readout. This is why the empirical analysis reports both encoder radii R_E and rollout radii R_F.

Proposition 2 (ACPC controls candidate-cost drift)**.** *Assume that* J *is locally* L_J*-Lipschitz on the clean/perturbed rollout-readout neighborhood evaluated by the fixed candidate set under metric* d_H*. If*

$$d_H\big(\Pi(F_\theta^{1:H}(E_\theta(h), \mathbf{a})), \Pi(F_\theta^{1:H}(E_\theta(\tilde{h}), \mathbf{a}))\big) \leq \epsilon,$$

then

$$|C_h(\mathbf{a}, g) - C_{\tilde{h}}(\mathbf{a}, g)| \leq L_J \epsilon.$$

Proof. The inequality is the local Lipschitz condition applied directly to the clean and perturbed rollout-readout sequences. □

In the reported basin diagnostic, this condition is instantiated with the L2 rollout-token distance underlying R_F; other downstream readouts are treated only as exploratory diagnostics.

Proposition 3 (Candidate top-1 stability)**.** *Let* $\mathcal{A} = \{\mathbf{a}^1, \ldots, \mathbf{a}^K\}$ *be the shared candidate set. Assume lower cost is preferred, and let* $\mathbf{a}^{(1)}$ *and* $\mathbf{a}^{(2)}$ *denote the lowest- and second-lowest-cost candidates on the clean branch. Suppose that for every candidate* j*,*

$$|C_h(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^j, g)| \leq \eta.$$

If the clean branch has a strict top-1/top-2 margin

$$\Delta = C_h(\mathbf{a}^{(2)}, g) - C_h(\mathbf{a}^{(1)}, g) > 2\eta,$$

then the perturbed branch selects the same top-1 candidate.

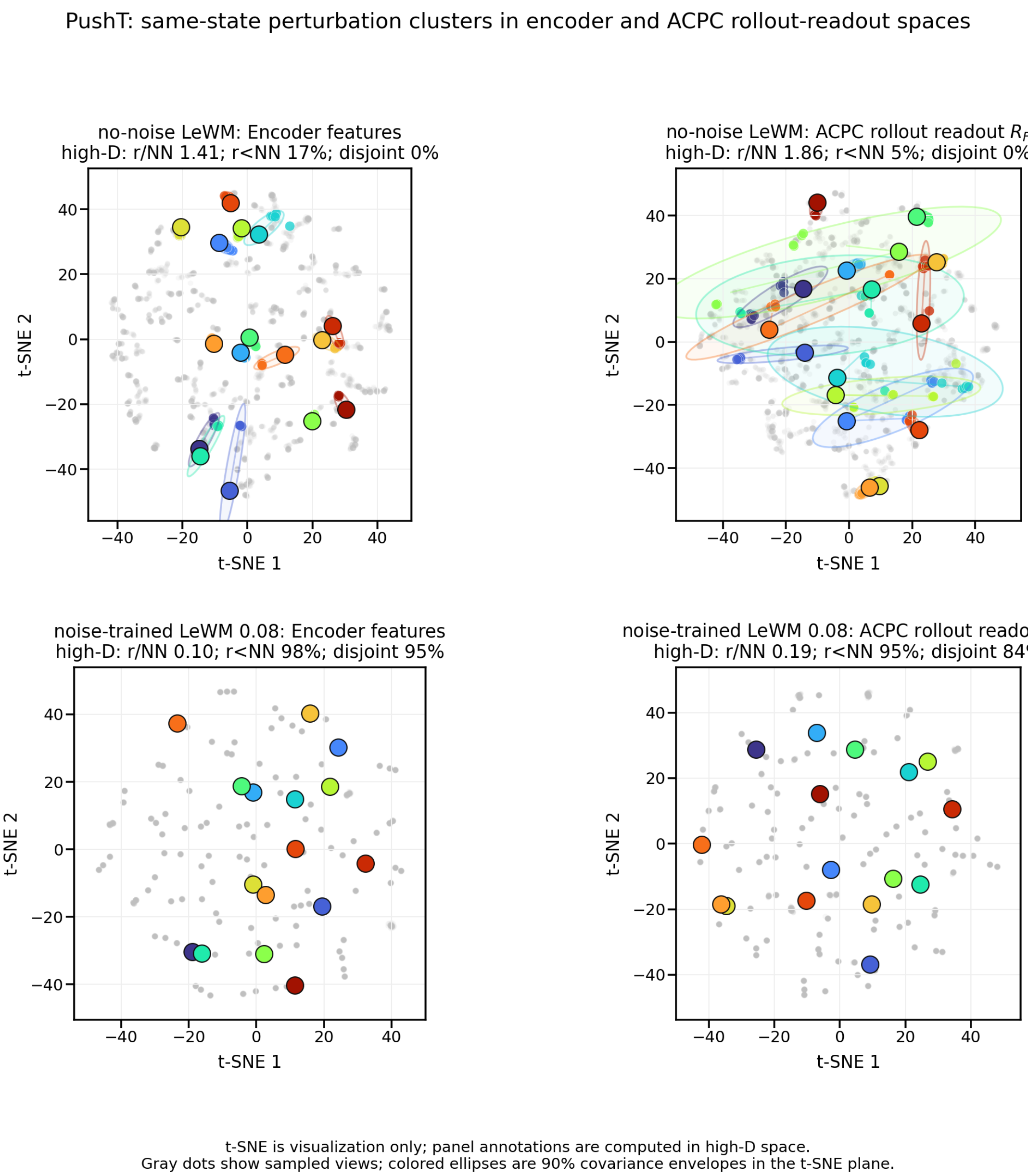

Figure 1: Empirical ACPC-basin visualization used as the paper's main visual reference. Top: no-noise-trained LeWM on PushT; bottom: noise-trained LeWM at $\sigma_{\max} = 0.08$. Left: encoder-history features; right: ACPC rollout readout R_F under the same recorded action sequence. t-SNE coordinates and envelopes are visualization only; small panel summaries report original-space local statistics. The quantitative multi-task basin summary is reported in Table 5.

Proof. For any candidate $\mathbf{a}^j \neq \mathbf{a}^{(1)}$,

$$C_{\tilde{h}}(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^{(1)}, g) \geq C_h(\mathbf{a}^j, g) - C_h(\mathbf{a}^{(1)}, g) - 2\eta \geq \Delta - 2\eta > 0.$$

Thus every non-top candidate remains more costly than the clean top-1 candidate on the perturbed branch. $\square$

Corollary 1 (ACPC-margin condition for candidate stability)**.** *Let $\mathcal{A} = \{\mathbf{a}^1, \ldots, \mathbf{a}^K\}$ be a shared candidate set and assume the cost readout J is locally L_J-Lipschitz as in Proposition 2. If every candidate in $\mathcal{A}$ has rollout-readout discrepancy at most ϵ between the clean and perturbed branches, and the clean branch has top-1/top-2 margin $\Delta > 2L_J\epsilon$, then the two branches select the same top-1 candidate from $\mathcal{A}$.*

Proof. Proposition 2 gives $|C_h(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^j, g)| \leq L_J\epsilon$ for every candidate. Applying Proposition 3 with $\eta = L_J\epsilon$ gives the result. $\square$

The corollary gives a formal link between ACPC and planning without turning the diagnostic into a closed-loop guarantee. It connects paired predictive agreement to candidate-cost stability and then to action selection under a margin condition. Candidate-ranking agreement and margin-conditioned action flips are downstream readouts of this chain. The result is limited to a fixed candidate set; CEM resampling, repeated replanning, and environment feedback can still change the closed-loop trajectory.

3.4 Sampled-pool stability and Gaussian sensitivity

The fixed-candidate statement is the deterministic core. A single MPC or CEM iteration, however, evaluates a sampled candidate pool. The next bound keeps the same diagnostic scope while exposing the two terms the empirical readouts estimate: ACPC tails and clean-margin tails.

Theorem 1 (Sampled-pool ACPC stability)**.** *Let $\mathcal{A} = \{\mathbf{a}^1, \ldots, \mathbf{a}^K\} \sim q^K$ be a once-sampled candidate pool, and use deterministic tie-breaking for* $\arg\min$. *For each sampled candidate define*

$$D_j = d_H\big(\Pi(F_\theta^{1:H}(E_\theta(h), \mathbf{a}^j)), \Pi(F_\theta^{1:H}(E_\theta(\tilde{h}), \mathbf{a}^j))\big).$$

Assume J is locally L_J-Lipschitz on the evaluated rollout-readout neighborhood and that a single draw from q satisfies

$$\Pr_{\mathbf{a}\sim q}[D(\mathbf{a}) > \epsilon] \leq \delta.$$

Let $\Delta_{\mathcal{A}} = C_h(\mathbf{a}^{(2)}, g) - C_h(\mathbf{a}^{(1)}, g)$ be the clean branch top-1/top-2 margin inside the sampled pool. Then

$$\Pr_{\mathcal{A}}\left[\arg\min_j C_h(\mathbf{a}^j, g) \neq \arg\min_j C_{\tilde{h}}(\mathbf{a}^j, g)\right] \leq K\delta + \Pr_{\mathcal{A}}[\Delta_{\mathcal{A}} \leq 2L_J\epsilon].$$

Theorem 1 is not a convergence theorem for adaptive multi-round CEM. It controls the instability of one sampled candidate pool. The first term is the probability that at least one candidate has a large clean/noisy rollout discrepancy; the second term is the probability that the clean candidate pool has too small a top-1 margin. This is why the shared-candidate diagnostics are reported as a family rather than a single scalar: ACPC-H and PCC estimate rollout/cost drift, CRA estimates ranking stability, and MAF estimates the high-margin flip failure term. A repeated replanning episode simply accumulates such risks unless additional assumptions are made.

Corollary 2 (Replanning union bound)**.** *If replanning step t has sampled-pool flip-probability bound*

$$r_t = K_t\delta_t + \Pr[\Delta_{\mathcal{A}_t} \leq 2L_J\epsilon_t],$$

then over T replanning steps,

$$\Pr[\exists t \leq T: \ \textit{the sampled-pool top-1 flips at step } t] \leq \sum_{t=1}^{T} r_t.$$

Proposition 4 (Finite-sample tail calibration for paired diagnostics)**.** *Fix the diagnostic sampling distribution used to draw independent paired clean/noisy histories and candidate action sequences. Let*

$$X_i = \mathbf{1}[D_i > \epsilon], \qquad i = 1, \dots, n,$$

where D_i is the paired ACPC rollout-readout discrepancy for diagnostic item i. Let $p_\epsilon = \Pr[D > \epsilon]$ and $\hat{p}_\epsilon = n^{-1} \sum_i X_i$. Then, with probability at least $1 - \delta$ over the diagnostic sample,

$$p_\epsilon \le \hat{p}_\epsilon + \sqrt{\frac{\log(1/\delta)}{2n}}.$$

The same one-sided calibration applies to any bounded Bernoulli readout computed from paired candidate costs, such as a margin-conditioned action-flip indicator.

This calibration is with respect to the released diagnostic sampling distribution, its independent-sample idealization, and the finite candidate construction. It does not convert an empirical median, quantile, or flip-rate table into a uniform bound over all CEM samples or all closed-loop states; when nearby dataset windows are dependent, we use it only as calibration.

The Gaussian stressor also has a local sensitivity interpretation. Let $G_{\mathbf{a}}(z) = \Pi(F_\theta^{1:H}(z, \mathbf{a}))$ and let $\tilde{o} = o + \xi$ with $\xi \sim \mathcal{N}(0, \sigma^2 I)$.

Proposition 5 (Local Gaussian ACPC sensitivity)**.** *If E_θ and $G_{\mathbf{a}}$ are locally twice differentiable around o and $E_\theta(o)$ with bounded second-order remainders, then for small σ,*

$$G_{\mathbf{a}}(E_\theta(o + \xi)) - G_{\mathbf{a}}(E_\theta(o)) = J_{G_{\mathbf{a}}}(E_\theta(o)) J_E(o) \xi + R_\xi,$$

where the remainder contributes higher-order terms, and

$$\mathbb{E}_\xi \left[\| G_{\mathbf{a}}(E_\theta(o + \xi)) - G_{\mathbf{a}}(E_\theta(o)) \|_2^2 \right] = \sigma^2 \, \| J_{G_{\mathbf{a}}}(E_\theta(o)) J_E(o) \|_F^2 + O(\sigma^3).$$

Thus Gaussian ACPC measures action-conditioned encoder–predictor sensitivity, not encoder invariance alone. Small encoder shifts can still be amplified by $J_{G_{\mathbf{a}}}$, while large nuisance shifts can become harmless after rollout if the product $J_{G_{\mathbf{a}}} J_E$ is small in those directions. The selective target is therefore asymmetric: the product should be small along nuisance perturbation directions, while rollout readouts for action-relevant state differences remain separated. The task-grounded near-boundary proxy margin pass event used later is

$$M = \mathbf{1}[d_{\text{proxy diff}} > d_{\text{same state noise}} + \delta_m] \, ,$$

with $\delta_m = 0$ in the reported table. It is a finite state-proxy selective-consistency check, not a full semantic-label coverage guarantee.

The theoretical statements in this section are intended as diagnostic calibration rather than as closed-loop control guarantees. They identify the failure modes that a paired clean/noisy diagnostic should measure—rollout-disagreement tails and clean candidate-margin tails—and explain why the Gaussian basin radius is an action-conditioned encoder–predictor sensitivity measurement. They do not assert that a low empirical median radius is a uniform bound for adaptive CEM, repeated replanning, or environment-feedback stability.

Selective ACPC pseudo-metric. For a nonempty fixed candidate family $\mathcal{A}$, the stability condition can be summarized by

$$D_{\text{ACPC}}^{\mathcal{A}}(h, \tilde{h}) = \max_{\mathbf{a} \in \mathcal{A}} d_H \big(\Pi(F_\theta^{1:H}(E_\theta(h), \mathbf{a})), \Pi(F_\theta^{1:H}(E_\theta(\tilde{h}), \mathbf{a})) \big) \, . \tag{8}$$

When d_H is a metric (or a pseudo-metric, if some horizon weights vanish), $D_{\text{ACPC}}^{\mathcal{A}}$ is a pseudo-metric on histories after passing through the encoder, predictor, and readout: distinct histories can have zero distance if all candidate rollouts are readout-equivalent. This follows because each map $h \mapsto \Pi(F_\theta^{1:H}(E_\theta(h), \mathbf{a}))$ pulls back d_H to a pseudo-metric, and the pointwise maximum of pseudo-metrics is again a pseudo-metric. The useful object is therefore *selective* predictive stability. Small $D_{\text{ACPC}}^{\mathcal{A}}$ for same-state visual perturbations gives candidate-cost and top-1 stability under the preceding Lipschitz and margin assumptions; the same contraction is not desirable for state/action pairs whose external task readouts differ, which is why the discriminability guard is part of the definition rather than an auxiliary aesthetic check.

Corollary 3 (Fixed-candidate predictive stability)**.** *Assume $D^{\mathcal{A}}_{\mathrm{ACPC}}(h, \tilde{h}) \le \epsilon$ for a same-state visual-perturbation pair and that J is locally L_J-Lipschitz on the rollout readout. If the clean candidate margin satisfies $\Delta > 2L_J\epsilon$, then the clean and perturbed branches select the same top-1 candidate from the fixed set $\mathcal{A}$.*

Proof. By definition of $D^{\mathcal{A}}_{\mathrm{ACPC}}$, every candidate in $\mathcal{A}$ has rollout-readout discrepancy at most ϵ. Proposition 2 bounds the cost drift of every candidate by $L_J\epsilon$, and Proposition 3 gives top-1 stability when $\Delta > 2L_J\epsilon$. □

The statement becomes selective only when it is paired with a discriminability condition such as Equation (6). Without that second condition, fixed-candidate predictive stability can still be achieved by a collapsed encoder–predictor, as shown next.

Encoder closeness is not the right substitute for this condition. It is not necessary: a nuisance-subspace change can move $E_\theta(h)$ and $E_\theta(\tilde{h})$ far apart while leaving $\Pi(F^{1:H}_\theta(\cdot, \mathbf{a}))$ and the cost unchanged. It is not sufficient either: a small encoder displacement can flip a low-margin cost ranking when the predictor or cost readout has high local sensitivity.

Proposition 6 (ACPC alone permits collapse)**.** *There exist encoder–predictor pairs for which* $\mathrm{ACPC}_H = 0$ *for every same-state perturbation pair and every action sequence, while action-distinct states are mapped to identical rollout readouts.*

Proof. Let $E_\theta(h) = z_0$ for every history and let $F^k_\theta(z_0, \mathbf{a}) = u_0$ for every horizon k and action sequence $\mathbf{a}$. Then clean and perturbed branches always have identical rollout readouts, so same-state ACPC is zero. However, any two states that require different actions, costs, or transitions also have identical readouts, violating the discriminability condition in Equation (6). □

Thus low ACPC is meaningful only with a guard on action-relevant separation. This is the role of the transition-resolution and inverse-dynamics probes in the main diagnostics, and it is the failure exposed by the heteroscedastic-loss ablation in Appendix F. This is where ACPC differs from existing nuisance-feature analyses of latent prediction: those results help explain why latent prediction may suppress irrelevant or noisy features at the representation level, whereas the condition above asks whether the remaining perturbation changes action-conditioned rollout readouts and candidate costs while preserving action-relevant separations.

4 Study protocol

4.1 Models and noise intervention

LeWM [26] is an end-to-end JEPA world model trained with a latent prediction loss and SIGReg regularization; inference uses CEM for latent-space MPC. PushT (2D pushing), TwoRoom (2D navigation), Reacher (2D arm), and Cube (3D manipulation) form a heterogeneous stress panel spanning different nuisance/discriminability regimes. We add per-frame Gaussian noise to the input sequence as a full-sequence augmentation: each frame is independently noised with probability $p = 1.0$, and the standard deviation is drawn from $\mathrm{Uniform}(0, \sigma_{\max})$. Because the target future frame is also encoded from the noised sequence, this intervention is not an original-target denoising objective. We sweep $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$ and evaluate observation-only Gaussian noise with the goal image unperturbed as the primary robustness endpoint. Observation+goal Gaussian noise is retained as an auxiliary stress condition.

4.2 Evaluation and diagnostics

The canonical LeWM development grid contains 36 epoch-10 checkpoints for training seed 3072: four tasks and, for each task, a no-noise baseline plus eight noise-trained checkpoints. Each checkpoint is evaluated with three evaluation seeds (42/43/44) and 100 trajectories per seed; canonical tables report the population mean ± population std across those three evaluation seeds. A completed Gaussian training-seed replication adds independent LeWM training seeds 3073 and 3074 under the same nine-point Gaussian sweep and evaluation protocol. Together, seed 3072 plus replication seeds 3073/3074 provide three training seeds for the main Gaussian closed-loop endpoint in Table 3.

The main results follow the evidence chain used in the claims. Closed-loop tables report training-seed mean/std for Gaussian robustness. The frozen diagnostic validation then asks whether an ACPC-family score fixed on seed 3072 localizes held-out Gaussian plateaus on seeds 3073/3074. The margin-conditioned flip audit connects the sampled-pool bound to top-1 candidate stability, the reduced-budget CEM trace checks whether the same direction appears inside the actual optimizer loop, and the task-grounded near-boundary proxy margin checks selective discriminability. Unseen-stressor scores, PLDM replication, target-view and heteroscedastic-loss ablations, latent-noise probes, exact key names, and full diagnostic tables remain scope checks or reproducibility appendices.

PLDM replication is reported in Appendix H.

5 Experiments

The experiments use additive Gaussian pixel noise as the primary controlled stressor for a representation-diagnostic question. We first show that the stressor changes closed-loop behavior. We then ask where recovery appears in the model, prioritizing cross-task ACPC/paired-rollout evidence and keeping task-specific representation measurements only as discriminability checks. The study is a matched-stressor trace through the encoder–predictor–planner chain; point-best noise-level ranking, external method comparison, and broad perturbation-family transfer are outside its scope.

All results use the protocol in Section 4: LeWM-base denotes the no-noise checkpoint, and LeWM+noise denotes the eight-level $\sigma_{\max}$ sweep.

5.1 JEPA control fragility under Gaussian observation noise

Table 1 reports LeWM-base success rates under unperturbed and noised evaluation. Each training-seed value is first averaged over the three evaluation seeds; the table reports mean $\pm$ population std across the three training seeds.

Table 1: LeWM-base under test-time Gaussian-noise settings. Values are mean $\pm$ population std across three training seeds; each training-seed value averages three evaluation seeds $\times$ 100 trajectories. Drop is computed per training seed as clean success minus observation-noise 0.08 success, then summarized.

Task	unpert.	obs 0.05	obs 0.08	obs+goal 0.08	drop
TwoRoom	94.67 ± 0.72	81.33 ± 6.77	68.78 ± 2.45	53.56 ± 2.57	25.89 ± 2.38
PushT	81.78 ± 4.89	19.22 ± 9.12	7.22 ± 5.81	4.89 ± 1.64	74.56 ± 5.55
Reacher	59.22 ± 1.03	33.56 ± 1.77	18.22 ± 0.42	13.78 ± 1.50	41.00 ± 1.44
Cube	65.22 ± 1.10	46.67 ± 1.96	43.11 ± 3.27	45.78 ± 0.79	22.11 ± 2.78

LeWM-base is strong on unperturbed images, but observation noise alone already produces large closed-loop drops: PushT loses 74.56 ± 5.55 pts at observation-noise 0.08, Reacher 41.00 ± 1.44 pts, TwoRoom 25.89 ± 2.38 pts, and Cube 22.11 ± 2.78 pts. The observation+goal column shows the stronger full-input-stream Gaussian-noise stress case; it further degrades TwoRoom and Reacher, but is not the primary robustness endpoint because the goal image is also perturbed. The drop pattern across tasks remains informative: PushT degrades most sharply, consistent with contact-heavy continuous control being sensitive to visual precision.

PLDM provides a second-family boundary on this failure mode. Without noise augmentation, PLDM drops sharply on PushT ($75.33\% \pm 3.68$ unperturbed to $18.33\% \pm 2.05$ at observation-noise 0.08, 57.00 pt) and TwoRoom ($96.67\% \pm 1.70$ to $67.00\% \pm 2.16$, 29.67 pt), while Reacher and Cube have much smaller no-noise-training gaps (2.33 and 4.33 pt; Table 14). PLDM Reacher is therefore a useful boundary case: its no-noise baseline is already high, so the LeWM Reacher clean lift below is read as a LeWM fixed-epoch stabilization phenomenon rather than a universal property of noise training. On the PLDM tasks with substantial cliffs, noise-trained rows recover observation-noise performance; Cube remains weakly responsive (Appendix H). These rows provide replication and boundary evidence for the same diagnostic story.

5.2 Noise augmentation supplies recovered checkpoints for diagnosis

Table 2 summarizes the sweep points needed for the main argument; the complete 8-level grid is released in the manifest-listed canonical evaluation artifact. All entries use the unified $n = 3$ seeds (42/43/44) $\times$ 100 trajectories protocol.

Table 2: LeWM noise-sweep summary. Values are success rate mean $\pm$ population std across evaluation seeds. The two point-best columns may refer to different training-noise levels; they summarize the sweep envelope, while Figure 2 and the manifest-listed canonical evaluation artifact provide the grid context.

Task	unpert. base	obs 0.08 base	best clean row	best obs0.08 row	Main reading
TwoRoom	94.00 ± 3.56	65.67 ± 1.25	98.33 ± 0.47 (0.08)	97.67 ± 0.94 (0.08)	large recovery; visually redundant regime
PushT	86.33 ± 2.36	4.33 ± 1.25	89.67 ± 1.70 (0.03)	89.00 ± 4.32 (0.08)	large recovery; contact-sensitive state
Reacher	58.67 ± 1.25	18.33 ± 2.05	86.00 ± 2.94 (0.06)	84.67 ± 5.19 (0.07)	recovery plus fixed-epoch clean lift
Cube	66.67 ± 2.62	47.00 ± 6.38	69.33 ± 0.47 (0.01)	68.33 ± 2.49 (0.03)	weaker recovery signal

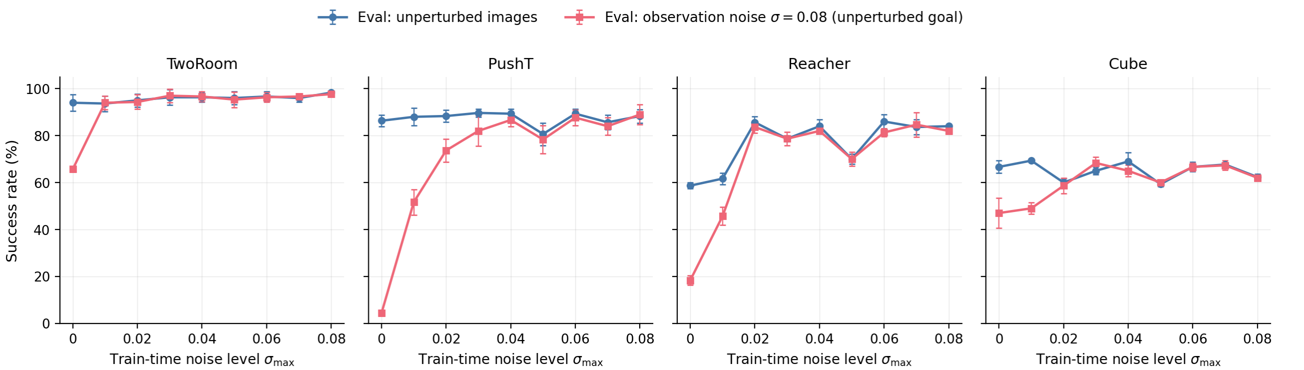

Figure 2: Noise-training sweep across four tasks. Blue circles: unperturbed evaluation; red squares: observation-only Gaussian noise $\sigma = 0.08$ with an unperturbed goal. Error bars show the population standard deviation across the three evaluation seeds. The curves document the intervention grid and the representative recovered checkpoints used for downstream diagnostics.

The sweep contributes two observations to the main argument. First, input-side noise can be an effective intervention in this protocol: PushT recovers from 4.33% to 89.00% at observation-noise 0.08, and Reacher from 18.33% to 84.67%. Second, the grid supplies representative recovered checkpoints for the downstream diagnostic question: what changed in the encoder–predictor–planner chain?

Reacher also shows a fixed-epoch clean-control lift. Noise training improves not only observation-noise robustness but also unperturbed control: the representative clean row rises from 58.67% at the no-noise baseline to about 86% under noise training. We interpret this as a fixed-protocol regularization or stabilization effect rather than pure robustness recovery. The diagnostic table supports this reading: Reacher does not show a resolution collapse under the fixed $\sigma_{\text{max}} = 0.08$ diagnostic checkpoint, with effective rank and transition resolution flat or slightly improved, while multi-step rollout drift drops sharply ($15.17 \to 0.36$). Thus the same intervention that contracts same-state noisy rollouts coincides with more stable clean action-conditioned prediction at epoch 10. This is a diagnostic observation, not a claim about asymptotic training behavior. The three-training-seed replication below makes the endpoint less dependent on seed 3072, but longer learning curves would still be needed to separate regularization from optimization timing.

The completed Gaussian training-seed replication strengthens the closed-loop part of the claim. It repeats the full Gaussian sweep for LeWM seeds 3073 and 3074 and combines them with the canonical seed-3072 sweep. Table 3 summarizes the observation-only Gaussian endpoint across all three training seeds.

Table 3: Three-training-seed Gaussian replication for LeWM. Values are mean ± population std across training seeds 3072/3073/3074; each seed point is itself the mean over evaluation seeds 42/43/44 with 100 trajectories per eval seed. The endpoint is observation-only Gaussian evaluation at 0.08 with the goal image unperturbed. Regret is best obs 0.08 minus the fixed $\sigma_{\max} = 0.08$ endpoint; the best-$\sigma_{\max}$ range shows the task/seed-dependent plateau.

Task	base obs 0.08	std0.08 obs 0.08	gain	best obs 0.08	regret	best $\sigma_{\max}$ range
TwoRoom	68.78 ± 2.46	97.11 ± 0.57	28.33 ± 2.68	97.67 ± 0.27	0.56 ± 0.79	0.05–0.08
PushT	7.22 ± 5.81	85.78 ± 3.45	78.56 ± 5.18	87.67 ± 0.98	1.89 ± 2.67	0.04–0.08
Reacher	18.22 ± 0.42	81.55 ± 0.88	63.34 ± 1.25	83.78 ± 0.63	2.22 ± 0.87	0.05–0.07
Cube	43.11 ± 3.27	62.56 ± 1.55	19.45 ± 4.53	66.44 ± 1.34	3.89 ± 2.20	0.03–0.05

Across three training seeds, the no-noise checkpoints retain large observation-noise cliffs, while the $\sigma_{\max} = 0.08$ endpoint recovers most of the best observed sweep performance. PushT remains the sharpest stress case, improving by 78.56 ± 5.18 points at the fixed high-noise endpoint; Reacher improves by 63.34 ± 1.25 points. The nonzero regret and task-dependent best-$\sigma_{\max}$ ranges support plateau wording rather than a universal optimum at exactly $\sigma_{\max} = 0.08$.

A three-seed blur/resize score check is reported in Appendix J. It is a deliberately narrow scope check on severe non-Gaussian stressors, not part of the three main results. The following validation stays within the frozen Gaussian grid and asks whether ACPC-family readouts identify plateau-member candidate regions across all three LeWM training seeds.

5.3 Three-seed frozen diagnostic validation

We freeze an aggregate diagnostic score on the seed-3072 development grid before evaluating independent training seeds: among nonzero training-noise checkpoints, lower ACPC-H/transition, lower PCC, higher CRA, and lower MAF receive better ranks, and the ranks are summed. We use this frozen score in two audit views. First, Table 4 reports a single-row proximity view: the minimum-score row in each task–seed block is judged only by distance to the closed-loop plateau. Second, Appendix L gives the main plateau-membership screen, which returns the top half of candidates in each task–seed block. No closed-loop success enters either diagnostic score.

The single-row proximity view is near the closed-loop plateau on held-out seeds without using their closed-loop scores: development seed 3072 gives 3/4 blocks within 5pp with 2.33 ± 3.49 pp regret; held-out seeds 3073/3074 give 7/8 blocks within 5pp with 2.21 ± 1.83 pp regret and a block-bootstrap 95% CI of $[1.04, 3.54]$; all seeds give 10/12 blocks within 5pp with 2.25 ± 2.51 pp regret. The task-level breakdown below is more useful for reading the failure modes than for ranking plateau-internal checkpoints.

Table 4: Three-seed frozen diagnostic validation on the full LeWM Gaussian grid. Within-5pp counts how often the single-row proximity view is within five percentage points of the closed-loop best observation-noise 0.08 checkpoint in the same task/seed block. Regret is best minus the reported row's observation-noise 0.08 success.

Task	exact best	within 5pp	regret to best	top-2 overlap
TwoRoom	1/3	3/3	0.56 ± 0.79	0.67/2
PushT	1/3	2/3	2.22 ± 2.47	0.67/2
Reacher	0/3	3/3	1.67 ± 0.94	1.67/2
Cube	0/3	2/3	4.56 ± 3.00	0.00/2
All	2/12	10/12	2.25 ± 2.51	0.75/2

Closed-loop evaluation establishes behavior; the frozen diagnostic screen tests whether ACPC-family readouts localize candidate regions that have entered a robustness plateau. Because the sweep has broad finite-variance plateaus, Appendix L treats $\sigma_{\max}$ as a candidate label inside each task–training-seed block and evaluates membership screens rather than point-best ranking. With a 5pp plateau label, 42 of the aggregate screen's 48 rows are plateau members (87.5% precision, 61.8% recall), matching the high-std top-half reference and trailing MAF-only, but above exact random top-half precision (70.8%). Presence is saturated, so the result supports plateau-entry enrichment and boundary localization; plateau-internal ordering is not a claim.

5.4 Paired Gaussian-noise ACPC basin diagnostic

We compute an eval-only ACPC basin diagnostic on the same 36 LeWM epoch-10 checkpoints used in the sweep. For each checkpoint we sample 100 fixed dataset windows and create eight paired same-state views using Gaussian pixel noise with std $\in \{0.01, \ldots, 0.08\}$, matching the training sweep's noise family. Each original/noised branch is encoded and then rolled out under the same recorded action sequence for horizon $H = 8$. The reported prediction readout is the identity on the resulting inference/cost latent rollout sequence, with L2 distance over rollout tokens. The released source and rendering script are listed in `DATA_MANIFEST.md`. The diagnostic intentionally excludes blur/resize so that the ACPC evidence is not confounded by train/eval perturbation-family mismatch.

For each checkpoint, let R_E be the median pairwise spread among the original and noised encoder-history views, normalized by the original-view local NN L2 scale, and let R_F be the median pairwise spread among their 8-step rollout latents, normalized by the unperturbed transition L2 reference over the same horizon. Lower R_F means the same-state perturbation views induce a tighter action-conditioned predictive basin; R_F/R_E is a descriptive contraction ratio rather than a success predictor.

This comparison localizes where the intervention changed same-state perturbation behavior. The compact rows summarize the high-noise regime visible in closed-loop evaluation; the frozen diagnostic validation in Table 4 audits plateau localization across the full three-seed grid. The full basin grid is released as `assets/paper1_data/acpc_basin_diagnostics.json`, and Section 5.5 reports a compact shared-candidate downstream check.

Table 5: LeWM Gaussian-noise ACPC-basin proxy: no-noise baseline vs. representative high-noise grid rows. R_E is normalized encoder-view spread and R_F is normalized action-conditioned rollout-view spread under the same action sequence. 8-step drift is the standard 8-step rollout-drift diagnostic at the same checkpoint; it is a rollout-space companion to R_F, not a discriminability guard. Drop is clean success minus observation-noise 0.08 success.

Task	checkpoint	obs 0.08	drop	R_E	R_F	8-step drift
TwoRoom	base	65.67	28.33	3.538	0.785	18.62
TwoRoom	noise 0.08	97.67	0.67	0.235	0.028	0.66
PushT	base	4.33	82.00	1.727	1.543	18.65
PushT	noise 0.08	89.00	−0.67	0.137	0.088	1.06
Reacher	base	18.33	40.33	4.238	1.012	15.17
Reacher	noise 0.07	84.67	−1.00	0.135	0.027	0.40
Cube	base	47.00	19.67	1.343	0.957	20.20
Cube	noise 0.03	68.33	−3.33	0.238	0.115	7.53

The compact rows quantify the characteristic basin movement illustrated in Figure 1: R_F is much smaller than at the no-noise baseline on the tasks with large recovery, with PushT decreasing from 1.543 to 0.088 and TwoRoom from 0.785 to 0.028. The companion 8-step rollout-drift column gives the same rollout-space localization in the standard diagnostic suite: it drops sharply on TwoRoom, PushT, and Reacher, while Cube's weaker recovery leaves a larger residual drift. Appendix B.8 adds a projection-free local-atlas check. The comparison localizes a joint contraction pattern after the closed-loop sweep: recovered checkpoints usually shrink both R_E and R_F, with 8-step drift as a rollout-stability companion; the anti-collapse evidence comes from the downstream discriminability checks.

The basin contraction is not treated as sufficient evidence of robustness. It is paired with the downstream readouts in Table 6 and the discriminability checks in Table 8: the former checks that shared-candidate costs and rankings move in the expected direction, while the latter rules out the trivial explanation that low R_F was obtained by erasing action-relevant state distinctions.

5.5 Downstream shared-candidate readouts move consistently

The fixed-candidate stability argument predicts that smaller same-action rollout disagreement should reduce downstream candidate-cost drift and ranking instability, subject to the stated margin assumptions. Table 6 reports a compact LeWM view from the released paired-diagnostic run using the primary observation-noise endpoint, horizon $H = 8$, and shared candidate-action sets. These representative rows connect the basin diagnostic to planner-side quantities; Table 4 gives the separate three-seed frozen-score readout.

Table 6: Compact LeWM shared-candidate readouts at the primary observation-noise endpoint. Lower is better for ACPC-H/transition, PCC, and MAF; higher is better for CRA. The full LeWM/PLDM readout artifact is listed in `DATA_MANIFEST.md`.

Task	obs-noise success	ACPC-H/trans.	PCC	CRA	MAF
TwoRoom	65.67 → 97.67	1.178 → 0.041	42.0 → 1.0	0.364 → 0.999	0.81 → 0.04
PushT	4.33 → 87.67	2.625 → 0.146	56.6 → 2.6	0.412 → 0.997	0.98 → 0.08
Reacher	18.33 → 83.67	2.092 → 0.029	72.2 → 0.5	0.278 → 1.000	0.91 → 0.01
Cube	47.00 → 67.33	1.979 → 0.058	63.7 → 1.3	0.274 → 0.999	0.89 → 0.00

These rows connect the ACPC basin to planner-facing quantities: on all four LeWM tasks, smaller paired rollout disagreement coincides with lower candidate-cost drift, higher candidate-ranking agreement, and fewer high-margin action flips. This is the empirical counterpart of Theorem 1: ACPC-H and PCC estimate the drift/tail side of the bound, CRA checks whether sampled candidate rankings remain aligned, and MAF estimates the high-margin flip failure term after margin filtering. The frozen diagnostic validation in Table 4 uses the same theory-matched readout family for no-retraining plateau localization.

Because the sampled-pool link depends on the clean top-1/top-2 candidate margin, we also run a sample-level margin-conditioned action-flip audit on the same fixed checkpoints. For each sampled state we record the clean candidate margin and whether the clean and noisy branches choose different top-1 candidates from the same 65-action candidate set. Table 7 reports flip rates after filtering to the top clean-margin quartile within each checkpoint; the all-sample column is shown as context. The recovered endpoint removes high-margin flips in all 12 task–training-seed blocks at this threshold. This is a finite fixed-pool calibration of the clean-margin term, not a closed-loop CEM guarantee.

Appendix M narrows the remaining planner gap with a reduced-budget offline CEMSolver trace on the same three seeds and base/std0.08 endpoints. Under matched clean/noisy histories and the same solver seed, the fixed high-noise endpoint lowers final plan L2 per dimension on all tasks (for example PushT 1.96 → 0.64, Reacher 1.15 → 0.38) and sharply reduces final best-cost deltas. Seeded top-1 flips are also lower on PushT, Reacher, and Cube, but remain high on TwoRoom after CEM updates; this is planner-side evidence for reduced optimizer sensitivity, not a complete CEM stability claim.

Table 7: Sample-level margin-conditioned action-flip audit over training seeds 3072/3073/3074. The q75 columns filter to the top clean-margin quartile within each checkpoint (25 samples per task–seed–endpoint); all-sample columns use all 100 samples. Lower flip rate is better.

Task	q75 flip base	q75 flip std0.08	all flip base	all flip std0.08
TwoRoom	0.67 ± 0.07	0.00 ± 0.00	0.80 ± 0.03	0.03 ± 0.00
PushT	0.92 ± 0.06	0.00 ± 0.00	0.92 ± 0.03	0.03 ± 0.02
Reacher	0.79 ± 0.02	0.00 ± 0.00	0.88 ± 0.03	0.01 ± 0.00
Cube	0.84 ± 0.09	0.00 ± 0.00	0.92 ± 0.03	0.00 ± 0.01

5.6 Selective consistency and failure checks

Low ACPC would be uninformative if it came from collapsed representations. Because 8-step drift already appears in the ACPC-basin section as rollout-space localization, Table 8 keeps only the compact representation checks: effective rank, transition-resolution ratio, and inverse-dynamics probe R^2. Table 9 then adds the task-grounded near-boundary proxy margin readout.

Table 8: Compact representation discriminability checks: LeWM-base → the fixed $\sigma_{\max} = 0.08$ noise-trained checkpoint. These checks are paired with the task-grounded near-boundary proxy margin pass-rate in Table 9.

Task	$\sigma_{\max}$	eff. rank	trans. L2	ID probe R^2
TwoRoom	0.08	47.60 → 37.69	0.7216 → 0.6621	0.2889 → 0.1419
PushT	0.08	76.42 → 78.08	0.3015 → 0.2789	0.7739 → 0.7647
Reacher	0.08	61.04 → 66.20	0.3704 → 0.3831	0.1621 → 0.1767
Cube	0.08	73.25 → 74.97	0.4847 → 0.5085	0.6657 → 0.6342

The representation checks are paired with a task-grounded near-boundary proxy margin pass-rate. For each task, we derive a simple programmatic label from logged task state: TwoRoom agent doorway/room side, PushT T-block pose cell, Reacher target/end-effector relation quadrant plus distance bin, and Cube cube-pose/goal-relation cell. Each anchor searches inside the closest 35% state-distance neighborhood and selects the closest neighbor crossing this task-state label. A sample passes if this proxy-different clean rollout distance is larger than its same-state clean/noisy rollout radius. This is still a programmatic state-proxy audit, not an oracle hand-labeled semantic proof, but it is stricter than the earlier global distance sanity pass. Table 9 reports the three-training-seed mean/std for the no-noise baseline and the fixed $\sigma_{\max} = 0.08$ endpoint.

Table 9: Task-grounded near-boundary proxy selective ACPC study over training seeds 3072/3073/3074. Pass-rate is the fraction whose label-crossing proxy-different clean rollout distance exceeds its same-state noisy radius under the same-action rollout readout. Pairs are selected as closest task-label-crossing neighbors inside the closest 35% state-distance neighborhood.

Task	std	pass-rate	same radius	proxy diff	margin	local state
TwoRoom	0.0	0.34 ± 0.05	17.16 ± 0.60	15.39 ± 0.11	-2.16 ± 0.57	0.51
TwoRoom	0.08	0.99 ± 0.01	0.77 ± 0.09	15.80 ± 0.51	15.27 ± 0.54	0.51
PushT	0.0	0.44 ± 0.17	18.39 ± 1.24	17.59 ± 0.21	-0.58 ± 1.49	1.21
PushT	0.08	1.00 ± 0.00	0.70 ± 0.06	17.54 ± 0.06	16.27 ± 0.05	1.21
Reacher	0.0	0.73 ± 0.03	15.44 ± 0.51	18.79 ± 0.07	2.71 ± 0.38	1.36
Reacher	0.08	1.00 ± 0.00	0.27 ± 0.02	18.82 ± 0.08	18.50 ± 0.06	1.36
Cube	0.0	0.45 ± 0.03	19.33 ± 0.15	19.08 ± 0.12	-0.27 ± 0.12	3.90
Cube	0.08	1.00 ± 0.00	0.49 ± 0.05	19.24 ± 0.06	18.69 ± 0.13	3.90

The task-grounded near-boundary proxy margin supplies the missing selective half of ACPC more directly than rank or ID probes alone, while staying below an oracle semantic-label claim. The high-noise endpoint keeps label-crossing proxy-different rollouts separated while shrinking same-state noisy rollouts on all four tasks; the no-noise baselines are weaker under this stricter audit, with pass-rates as low as 0.34 on TwoRoom and 0.44 on PushT. Each task/endpoint aggregates three training seeds and 61–100 label-crossing anchors per seed. Treating these local pairs as independent only for calibration, Hoeffding gives $p \geq \hat{p} - \sqrt{\log(1/\alpha)/(2n)}$; for the smallest high-noise row (TwoRoom, 183 total anchors, $\hat{p} = 0.989$) this gives a one-sided lower calibration near 0.90, and for 300-anchor rows near 0.93. Nearby dataset windows may be dependent, so we use this as finite-sample calibration rather than a dataset-wide formal guarantee. The heteroscedastic-loss ablation in Appendix F remains the failure case that motivates the guard: PushT transition L2 falls from 0.3015 to 0.1023, the ID probe falls from 0.7739 to 0.2678, and unperturbed success collapses to 13.33%. The longer representation, rollout-drift, latent-noise, and task-by-task mechanism diagnostics remain in Appendix C–G.

5.7 Bounded unseen-stressor scope check

The main claim remains matched Gaussian-noise robustness. After freezing the Gaussian grid, we also ran a strongest-severity unseen-stressor check to test whether the diagnostic direction is specific outside the training perturbation family. Table 10 shows a split pattern. TwoRoom and Reacher exhibit clear positive score deltas under strongest blur, and the paired ACPC/PCC/CRA diagnostics move in the same direction. PushT and Cube do not show a clear score gain at the scale of training-seed and evaluation variance, and their diagnostic deltas are correspondingly weak or mixed. The selected rows follow the appendix convention: TwoRoom/Reacher blur are clear positive-transfer endpoints, while PushT/Cube resize are small-effect boundary endpoints. A full appendix audit over all task–family–seed endpoints for blur and resize gives the same directional readout: the composite ACPC/PCC/CRA/MAF rank has Spearman $\rho = 0.94$ with stress-success delta over 24 rows. We therefore use this table only as bounded scope evidence: ACPC-style paired diagnostics localize clear out-of-family improvements when present, but the current evidence does not justify a universal cross-perturbation robustness claim.

Table 10: Bounded unseen-stressor scope check. Score Δ is the fixed $\sigma_{\max} = 0.08$ checkpoint minus the no-noise baseline under the strongest unseen stress, averaged over training seeds 3072/3073/3074. Diagnostic deltas are from the matched paired-diagnostic slice averaged over training seeds 3072/3073/3074; negative is better for ACPC-H/transition and PCC, positive is better for CRA. The table is a scope check, not a universal transfer claim.

Task	stress	score Δ	Δ ACPC	Δ PCC	Δ CRA	reading
TwoRoom	blur $k = 15$	$+43.11 \pm 8.90$	-0.691	-44.7	$+0.616$	clear gain; diagnostics aligned
Reacher	blur $k = 15$	$+49.22 \pm 3.29$	-1.895	-48.9	$+0.587$	clear gain; diagnostics aligned
PushT	resize 0.25	$+2.89 \pm 17.98$	-0.267	-6.7	$+0.010$	no clear gain at this scale
Cube	resize 0.25	-0.89 ± 1.40	$+0.052$	-0.6	-0.030	no clear gain at this scale

6 Discussion and limitations

What the evidence supports. This is a controlled diagnostic study, not a method paper. The primary behavior statistic is the full three-training-seed Gaussian replication; evaluation-seed variance is kept as the within-seed measurement scale. These standard deviations are descriptive over three training seeds, not asymptotic significance claims, and correlated diagnostic windows are calibration evidence. Gaussian pixel noise remains the training/evaluation axis; blur/resize scores, PLDM replication, and failure-case ablations are scope checks. The frozen diagnostic plateau audit is a plateau-entry screen, so we do not claim point-optimal std prediction or checkpoint ranking inside a plateau. We do not claim superiority over DrQ-style augmentation, TD-MPC2, DreamerV3, or robust MPC methods; the contribution is a no-retraining diagnostic for JEPA latent world-model checkpoints under a controlled Gaussian stressor.

How to use the diagnostics. The readouts form an evidence chain rather than a single scalar. Closed-loop evaluation establishes the behavior. ACPC rollout/cost readouts localize predictive-dynamics change. Margin-conditioned flips and task-grounded near-boundary pairs check the clean-margin and selectivity side. The reduced-budget CEMSolver trace asks whether the same direction appears inside the optimizer loop without estimating the full adaptive CEM distribution or claiming adaptive CEM stability. The frozen diagnostic screen is a held-out plateau-membership audit. Cube remains the useful boundary case, so point-best $\sigma_{\max}$ values are plateau context rather than fine-grained ranking targets.

What remains. The data reject a too-strong reading of latent prediction: future-latent prediction alone does not guarantee closed-loop visual robustness. They also do not prove a universal cross-perturbation robustness claim. Stronger discriminability evidence would require hand-labeled or simulator-derived contact, topology, action-value, or cost-to-go labels, especially for PushT and Cube. The heteroscedastic NLL probe and target-view ablation bound the method story: prediction difficulty alone is the wrong gate, and perturbed-history $\to$ original-future targets are not sufficient here. A paired predictive-dynamics consistency objective remains the separate method-paper path after diagnostic validity is established.

7 Conclusion

This paper frames Gaussian-noise robustness in JEPA latent world-model control as a staged diagnostic problem. Encoder geometry shows how visual perturbations enter the latent state; ACPC asks whether they change action-conditioned futures and candidate costs; and a task-grounded near-boundary proxy margin checks whether robustness is bought by erasing coarse action-relevant distinctions. On fixed LeWM checkpoints, input-side noise training recovers broad matched-Gaussian robustness plateaus across three training seeds, and the paired diagnostics localize that recovery through action-conditioned rollout/cost stability, selective margins, and reduced optimizer sensitivity.

The theory is deliberately diagnostic: fixed-candidate ACPC bounds cost drift, sampled-pool ACPC exposes ACPC-tail and clean-margin failure modes, and local Gaussian linearization connects the measured basin radius to action-conditioned encoder–predictor sensitivity. The broader implication is not solved generalization; it is a concrete audit target for future methods: stable action-conditioned predictive dynamics under nuisance variation, while preserving distinctions needed for planning.

Acknowledgements

The public repository contains code, aggregate evaluation files, rendering scripts, and data/checkpoint pointers for this revision; JSON hashes are listed in the data manifest. We thank the LeWM authors for releasing the baseline framework on which this study builds.

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A Proofs and calibration for ACPC diagnostics

Reading. Diagnostic calibration; proofs follow.

Proof of Theorem 1. Let $\mathcal{E}$ be the event that every sampled candidate satisfies $D_j \le \epsilon$. By the single-candidate tail assumption and a union bound,

$$\Pr_{\mathcal{A}}[\mathcal{E}^c] \le \sum_{j=1}^{K} \Pr[D_j > \epsilon] \le K\delta.$$

On $\mathcal{E}$, Proposition 2 gives $|C_h(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^j, g)| \le L_J\epsilon$ for all candidates in the sampled pool. If also $\Delta_{\mathcal{A}} > 2L_J\epsilon$, Proposition 3 implies that the clean top-1 candidate remains top-1 on the perturbed branch. Therefore a flip can occur only if either $\mathcal{E}^c$ occurs or the sampled clean margin is at most $2L_J\epsilon$:

$$\Pr[\text{flip}] \le \Pr[\mathcal{E}^c] + \Pr[\Delta_{\mathcal{A}} \le 2L_J\epsilon] \le K\delta + \Pr[\Delta_{\mathcal{A}} \le 2L_J\epsilon].$$

Proof of Corollary 2. Let F_t be the event that the clean and perturbed branches select different top-1 candidates at replanning step t for the sampled pool at that step. The sampled-pool theorem gives $\Pr[F_t] \le r_t$ under the stated per-step assumptions. The probability of at least one flip is bounded by another union bound:

$$\Pr[\exists t \le T : F_t] \le \sum_{t=1}^{T} \Pr[F_t] \le \sum_{t=1}^{T} r_t.$$

No independence across replanning steps is required for this bound. It is intentionally loose and does not account for environment feedback after a flip.

Proof of Proposition 4. For a fixed threshold ϵ, the variables $X_i = \mathbf{1}[D_i > \epsilon]$ are independent Bernoulli variables under the diagnostic sampling distribution. The one-sided Hoeffding inequality gives

$$\Pr[p_\epsilon > \hat{p}_\epsilon + u] \le \exp(-2nu^2).$$

Setting $u = \sqrt{\log(1/\delta)/(2n)}$ gives the stated upper confidence bound. The same argument applies to any bounded Bernoulli diagnostic, including a margin-conditioned action-flip indicator, after fixing the diagnostic sampling distribution and candidate construction, with the same independence caveat.

Derivation of Proposition 5. By Taylor expansion of the encoder around o,

$$E_\theta(o+\xi) = E_\theta(o) + J_E(o)\xi + R_E(\xi), \qquad \|R_E(\xi)\| = O(\|\xi\|^2).$$

Expanding $G_{\mathbf{a}}$ around $E_\theta(o)$ and substituting the encoder expansion gives

$$G_{\mathbf{a}}(E_\theta(o+\xi)) - G_{\mathbf{a}}(E_\theta(o)) = J_{G_{\mathbf{a}}}(E_\theta(o))J_E(o)\xi + R_\xi,$$

with $\|R_\xi\| = O(\|\xi\|^2)$ under the bounded second-order remainder assumptions. For $\xi \sim \mathcal{N}(0, \sigma^2 I)$ and $A = J_{G_{\mathbf{a}}}(E_\theta(o))J_E(o)$,

$$\mathbb{E}\|A\xi\|_2^2 = \operatorname{tr}(A\,\mathbb{E}[\xi\xi^\top]A^\top) = \sigma^2\|A\|_F^2.$$

The cross and remainder terms are higher order because $\mathbb{E}\|\xi\|^3 = O(\sigma^3)$ and $\mathbb{E}\|\xi\|^4 = O(\sigma^4)$, yielding the stated expansion. This is a local small-noise calibration, not a finite-noise robustness theorem.

Hoeffding calibration for the task-grounded near-boundary proxy margin. Let $M_i \in \{0,1\}$ denote the pass event $M_i = \mathbf{1}[d_{\text{proxy diff},i} > d_{\text{same state noise},i} + \delta_m]$ for sampled pair i, and let $\hat{p} = n^{-1}\sum_i M_i$. Under independent-pair calibration, Hoeffding's inequality gives

$$\Pr[p < \hat{p} - \epsilon] \le \exp(-2n\epsilon^2),$$

so with probability at least $1-\alpha$,

$$p \ge \hat{p} - \sqrt{\frac{\log(1/\alpha)}{2n}}.$$

For the reported high-noise endpoints, each task/endpoint has $n = 300$ pairs after aggregating 100 pairs over three training seeds. A pass-rate $\hat{p} = 1.00$ gives $1 - \sqrt{\log 20/600} = 0.929$ at 95% confidence. Because adjacent dataset windows can be dependent, the main text uses this as calibration rather than as a dataset-wide guarantee.

One-step prediction-loss triangle. For one-step predictions, write $\hat{z}_{t+1} = F_\theta(E_\theta(h_t), a_t)$, $\hat{\tilde{z}}_{t+1} = F_\theta(E_\theta(\tilde{h}_t), a_t)$, $z_{t+1} = E_\theta(h_{t+1})$, and $\tilde{z}_{t+1} = E_\theta(\tilde{h}_{t+1})$. Any metric d satisfies

$$d(\hat{z}_{t+1}, \hat{\tilde{z}}_{t+1}) \le d(\hat{z}_{t+1}, z_{t+1}) + d(z_{t+1}, \tilde{z}_{t+1}) + d(\tilde{z}_{t+1}, \hat{\tilde{z}}_{t+1}).$$

Thus clean prediction error, noisy-branch prediction error, and target-future nuisance stability jointly upper-bound one-step predictive disagreement. This teacher-forced one-step inequality does not guarantee rollout stability or closed-loop robustness; the target-view ablation in Appendix N shows that perturbed-history to original-future denoising is not sufficient in this study.

B Experimental details

Reading. Reproducibility details.

B.1 Environments

- **PushT.** 2D continuous pushing; 20,000 expert episodes; $\sim$196 steps/episode; action dim 2 (direction + magnitude); 224×224 RGB.
- **TwoRoom.** 2D continuous navigation; 10,000 episodes; $\sim$92 steps; action dim 2; 224×224 RGB.
- **Reacher.** 2D arm control (DeepMind Control Suite); 10,000 episodes; 200 steps; action dim 2.
- **Cube.** 3D cube manipulation (OGBench); 10,000 episodes; 200 steps; action dim 7.

B.2 Noise augmentation implementation

The actual implementation is `utils.py::AddNormalizedGaussianNoise`. Key points: (i) noise is added on ImageNet-normalized tensors and must be divided by the per-channel std to retain "pixel-space std" semantics; (ii) sampling is per-frame independent over the leading frame dims, not per-batch; (iii) all sweeps in this paper fix the noise probability to 1.0, set the lower std to 0, and vary only the upper std $\sigma_{\max}$.

```
class AddNormalizedGaussianNoise:
    """Per-frame independent. Each frame draws Bernoulli(noise_prob) then
    std ~ Uniform(std_min, std_max). Pixel-space std is converted to
    normalized space by dividing by per-channel std before adding."""
    def __init__(self, std_min, std_max, noise_prob=1.0):
        self.std_low, self.std_high, self.noise_prob = std_min, std_max, noise_prob
        self.channel_std = torch.as_tensor(IMAGENET_STD) # (C,)

    def __call__(self, x):
        if self.std_high <= 0 or self.noise_prob <= 0:
            return x
        leading = x.shape[:-3]
        stds = torch.empty(leading, device=x.device, dtype=x.dtype).uniform_(
            self.std_low, self.std_high)
        if self.noise_prob < 1.0:
            mask = (torch.rand(leading, device=x.device) < self.noise_prob).to(x.dtype)
            stds = stds * mask
        per_frame_scale = stds.view(*leading, 1, 1, 1)
        channel_factor = (1.0 / self.channel_std.to(x.device, x.dtype)).view(
            *([1] * len(leading)), -1, 1, 1)
        scale = per_frame_scale * channel_factor # pixel-space std -> normalized
        return x + torch.randn_like(x) * scale
```

B.3 Evaluation protocol

Unperturbed evaluation uses no noise, 100 trajectories per seed × 3 seeds (42/43/44), totalling 300 trajectories per condition. The primary noisy evaluation adds Gaussian noise of std $\in \{0.05, 0.08\}$ to observation pixels while leaving the goal image unperturbed under the same 3-seed protocol. Observation+goal Gaussian noise is retained as an auxiliary full-input-stream stress condition and is reported separately where used. Success criterion is task-specific (PushT: T-block pose match; TwoRoom: target region; Reacher: joint-angle match; Cube: cube position match).

B.4 Full LeWM noise-sweep grid

The complete LeWM 4-task × 9-configuration sweep behind Table 2 and fig. 2 is released in `assets/paper1_data/canonical_evals_20260517.json` and indexed in `DATA_MANIFEST.md`. The PDF keeps the compact sweep summary and figure rather than duplicating the full grid tables.

B.5 Full LeWM ACPC-basin grid

The full 36-row LeWM ACPC-basin grid is released in `assets/paper1_data/acpc_basin_diagnostics.json`; the compact baseline-vs-recovered rows used for the argument are in Table 5. This keeps the PDF focused on the evidence needed to read the claim while preserving the full grid as an auditable artifact.

B.6 Compute

Training runs on a single NVIDIA H800 (80 GB) and follows the LeWM training schedule released with the baseline.

B.7 Main-figure rendering

Figure 2 is produced by the main plot script listed in `DATA_MANIFEST.md`; no checkpoint or model loading is required.

Figure 1 is produced by `tools/paper1_selective_contraction.py`. The rendering command used for this paper is:

```
python -m tools.paper1_selective_contraction \
  --plot-clusters --plot-tasks PushT \
  --n-sequences 128 --cluster-anchor-count 16 \
  --view-stds 0.0 0.01 0.04 0.08 \
  --cluster-perturb-repeats 6 \
  --feature-cache-dir /tmp/paper1_selective_contraction_cache \
  --cluster-out-dir assets/paper1_figs \
  --cluster-envelope ellipse --cluster-envelope-coverage 0.90
```

Unlike the aggregate figure generator, this optional rendering path loads the PushT checkpoints and dataset windows to draw repeated same-state perturbation samples. The feature-cache directory stores the extracted encoder and rollout feature arrays as parameter-keyed `.npz` files, so label/style-only redraws do not reload checkpoints. Use `--refresh-feature-cache` only when changing checkpoints, sampled windows, view stds, or feature-extraction parameters. The small panel summaries report high-dimensional statistics; the t-SNE ellipses are only qualitative 2-D covariance envelopes.

B.8 Local cluster atlas: a projection-free check of the basin visualization

Because t-SNE distorts global distances, Figure 3 provides a complementary *local normalized atlas* view of the same PushT data. Each small box selects one original dataset state; the circle radius equals the distance to its nearest *other* original state, computed in the original high-dimensional feature space, and the color markers are that state's original and Gaussian-perturbed views after a local PCA used only for in-box display. The visual question per box is therefore exactly the high-dimensional one: does the same-state perturbation cloud stay inside the nearest-different-state radius? On the no-noise-trained checkpoint many perturbation views escape their circles (median $r/\mathrm{NN} > 1$, few disjoint balls); on the full-sequence noise-trained checkpoint the clouds contract well inside the circles in both encoder features and the ACPC rollout readout R_F. This avoids the global-projection caveats of Figure 1 at the cost of showing only local neighborhoods; the quantitative multi-task evidence remains Table 5.

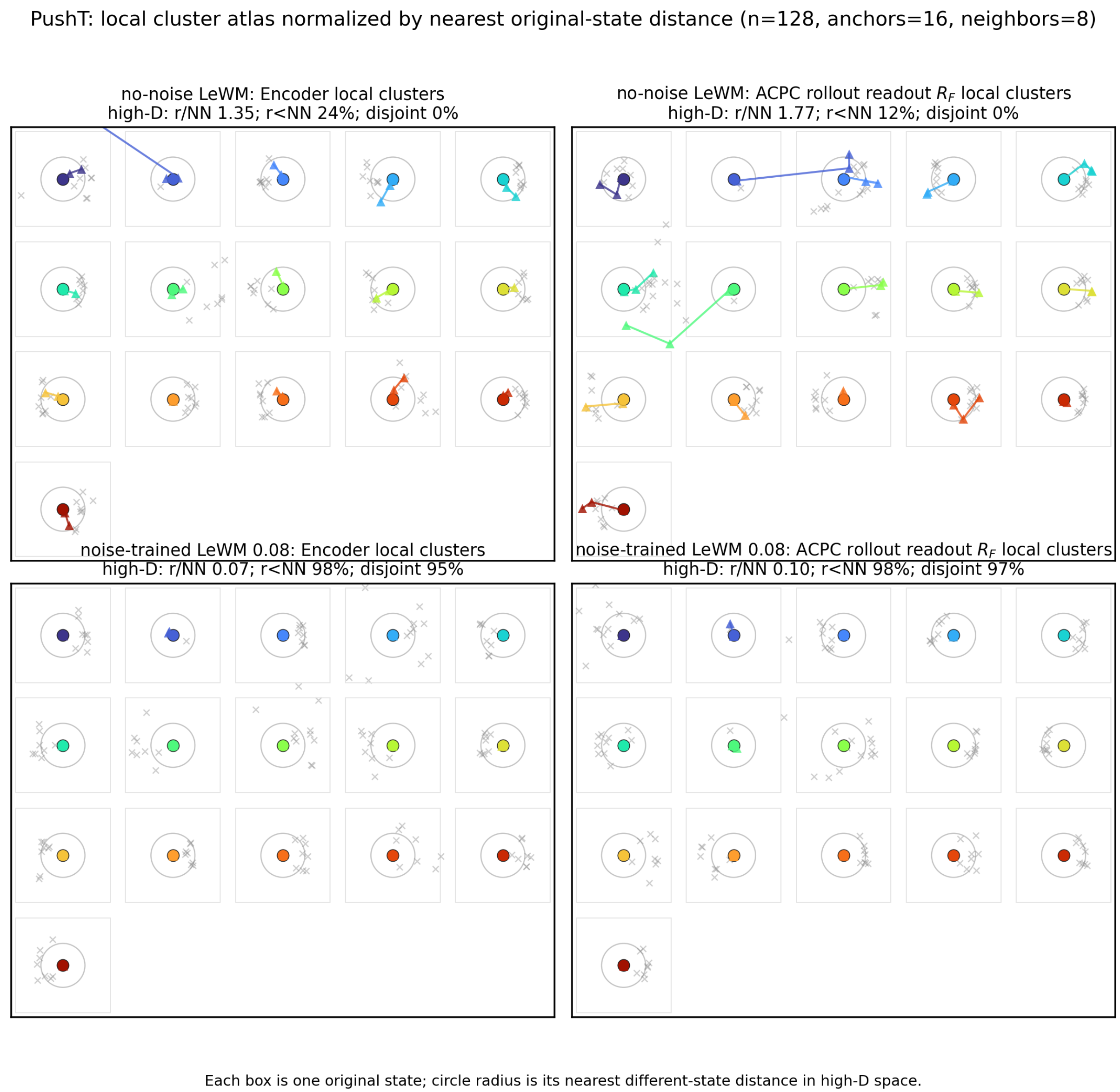

Figure 3: PushT local cluster atlas ($n = 128$ windows, 16 anchors, 8 neighbors). Each box is one original state with Gaussian-perturbed views after local PCA; the circle radius is the high-dimensional nearest-other-state distance. Panel summaries report original-space local statistics.

The atlas rendering command is:

```
python -m tools.paper1_selective_contraction \
  --plot-atlas --plot-tasks PushT \
  --n-sequences 128 --atlas-anchor-count 16 \
  --view-stds 0.0 0.01 0.04 0.08 \
  --feature-cache-dir /tmp/paper1_selective_contraction_cache \
  --atlas-out-dir assets/paper1_figs
```

B.9 Perturbation visualization assets

Representative Gaussian-noise and blur renderings for all four tasks are released under `assets/paper1_figs/corruption/`. The PDF omits these perturbation panels because the perturbation definitions and score tables determine the protocol.

C Diagnostic framework definitions

Reading. Auxiliary definitions.

C.1 LeWM training objective

LeWM [26] is trained with latent prediction and SIGReg regularization:

$$\mathcal{L}_{\text{LeWM}} \;=\; \mathcal{L}_{\text{pred}} \;+\; \lambda \cdot \mathcal{L}_{\text{SIGReg}}\,, \tag{9}$$

where $\mathcal{L}_{\text{pred}}$ is the latent-space MSE between predicted and target representations. SIGReg projects each latent onto unit-norm random directions, computes the Epps–Pulley empirical-characteristic-function distance [8] between each projection and $\mathcal{N}(0,1)$, and aggregates with Gauss-window weights to prevent collapse without explicit BatchNorm. Inference uses CEM [22] for latent-space MPC [9].

C.2 Five-layer diagnostic framework

The diagnostic framework decomposes the JEPA world-model control pass into pixels $\to$ encoder f $\to$ latent z $\to$ predictor g $\to$ cost surface $\to$ planned actions. Layers 1–2 describe whether visual perturbations enter the latent and how they change latent geometry. Layer 3 measures how the predictor amplifies an encoder shift. Layer 4 injects noise directly into z to separate encoder effects from predictor and cost effects. Layer 5 checks task-relevant information retained in the latent. We use nearest-neighbor distance as a local latent scale primitive in the spirit of kNN out-of-distribution detection [35]; the scale-normalized ratios are introduced for this diagnostic study.

Minimal notation. Let $z_i = f(x_i)$ and $\tilde{z}_i = f(\tilde{x}_i)$ be unperturbed and noised latents for the same token. The distance, local nearest-neighbor (NN) scale, and three introduced ratios are

$$\begin{aligned}
d_{\cos}(a,b) &= 1 - \frac{a^\top b}{\|a\|\,\|b\|}, \\
d_i^{\text{NN}} &= \min_{j\neq i} d_{\cos}(z_i, z_j), \\
r_i^{\text{enc}} &= \frac{d_{\cos}(z_i,\tilde{z}_i)}{d_i^{\text{NN}}+\epsilon}, \\
r_i^{\text{pred}} &= \frac{d_{\cos}\big(g(z)_i, g(\tilde{z})_i\big)}{d_i^{\text{NN}}+\epsilon}, \\
R_d &= \frac{\text{median}_t\ d(z_t, z_{t+1})}{\text{median}_{(t,t')\in\mathcal{F}}\ d(z_t, z_{t'})+\epsilon}.
\end{aligned} \tag{10}$$

Here r_i^{enc} is the encoder-shift ratio, r_i^{pred} is the fragility ratio, and R_d is the transition-resolution ratio for either cosine or L2 distance d. $\mathcal{F}$ denotes random far pairs from different temporal locations, and $\epsilon = 10^{-8}$ is a numerical-stability constant.

Layer metrics. Layer 1 reports raw angular/L2 encoder shift, angular gain per unit noise, and the encoder-shift ratio in Equation (10). Layer 2 reports local neighborhood scale, effective rank [31], and CKA [20]. Layer 3 reports fragility ratio and multi-step rollout L2 drift. Layer 4 reports latent-noise cost-surface slope and latent robust radius. Layer 5 reports the L2/cosine transition-resolution ratio and inverse-dynamics linear-probe R^2 [1]. All five layers are descriptive diagnostics used to localize what changed at evaluated checkpoints; none is treated as a standalone checkpoint selector.

D Full diagnostic profile of LeWM-base on four tasks

Reading. The full LeWM-base diagnostic profile is released in `assets/paper1_data/canonical_diagnostics_20260517.json`; reported key names are indexed in `DATA_MANIFEST.md`. The main text reports only the compact discriminability and rollout/cost readouts needed for the argument.

E Heteroscedastic-loss formulation

Reading. Ablation loss definition.

The scale-preserving heteroscedastic NLL referenced in Section 6 and detailed in Appendix F is:

$$\mathcal{L}_{\text{hetero}} = \tfrac{1}{2}\,\exp(-s_t)\,\|z_{t+1} - \hat{z}_{t+1}\|^2 + \tfrac{1}{2}\,s_t\,, \tag{11}$$

where s_t is the σ-head-predicted log-variance. During training $\exp(-s_t)$ acts as an automatic weight: high-error transitions are down-weighted and low-error transitions are up-weighted. When $s_t \equiv 0$ the loss reduces to plain MSE.

F Heteroscedastic-loss negative result (data)

Reading. This appendix records the full heteroscedastic-loss data referenced in Section 6. The σ-head is trained jointly with the predictor; gradient is detached on the σ path so the mean prediction path is exactly LeWM MSE when σ is constant.

Table 11: Heteroscedastic-loss evaluation (canonical 3-seed $\times$ 100 protocol). The LeWM+noise rows use representative observation+goal 0.08 high-noise rows (TwoRoom $\sigma_{\max} = 0.08$, PushT $\sigma_{\max} = 0.06$), matching the strongest stress column of this ablation; the observation-only representative rows in the canonical sweep artifact differ.

Task / model	unpert.	goal .05	obs .05	obs+goal .05	goal .08	obs+goal .08
TwoRoom LeWM-base	94.00	73.33	72.33	61.33	58.67	50.00
TwoRoom LeWM+noise row	98.33	98.00	98.33	98.00	98.67	98.67
TwoRoom hetero	99.67	85.33	96.67	84.67	73.33	55.33
PushT LeWM-base	86.33	38.00	17.00	12.00	11.33	4.67
PushT LeWM+noise row	89.33	87.67	88.00	88.33	89.67	87.00
PushT hetero	13.33	7.67	7.67	7.67	9.67	6.00

TwoRoom hetero reaches 99.67% on unperturbed evaluation — compatible with the prior that low-dimensional discrete tasks can tolerate stronger nuisance contraction / clustering — but lags noise training on high-noise robustness. **PushT hetero unperturbed success is** 13.33% — **a method-level failure**, not evidence for a useful robustness compromise.

Table 12: Representation diagnostics of the hetero-loss failure.

Metric	TwoRoom base	TwoRoom hetero	PushT base	PushT hetero
Local NN cosine distance	0.0449	0.0281	0.2360	0.1051
Effective rank	47.60	33.59	76.42	42.85
transition-resolution ratio (cos)	0.5538	0.3780	0.0868	0.0101
transition-resolution ratio (L2)	0.7216	0.6055	0.3015	0.1023
ID probe R^2	0.2889	−0.0573	0.7739	0.2678
Mean action-induced prediction shift	0.5329	0.4482	0.1283	0.0841
Multi-step rollout drift	18.62	17.90	18.65	14.01

Hetero-loss compresses the representation in both tasks. In TwoRoom (low-dimensional, discrete, redundant), the observed compression coexists with high unperturbed success, but the robustness shortfall shows that compression alone is not the target. For PushT, the L2 transition-resolution ratio collapses from 0.3015 to 0.1023 and ID probe R^2 from 0.7739 to 0.2678 — task-relevant state information is erased. The contrast with input-side noise training is sharp: the fixed $\sigma_{\max} = 0.08$ PushT checkpoint in Table 8 leaves rank and probe nearly unchanged, so this collapse is specific to error-based downweighting. The drop in multi-step rollout drift is not good news either: the latent has become more *predictable* without being more *controllable*.

Mechanism. The learned σ values align with visible high-error transition regimes (contact frames in PushT, doorway crossings in TwoRoom, etc.), but using σ as an automatic weight to down-weight high-error transitions misclassifies the contact-and-control-critical states of PushT as "unimportant" and erases them. This is a negative example of the broader selective-consistency lesson: in contact-heavy control, **hard $\neq$ unimportant**.

This finding motivates a narrower follow-up direction: per-token *consistency* controllers should be separated from loss reweighting, and any difficulty signal should be detached from the mean prediction path unless it is explicitly protected by an action-relevance guard. We leave that objective design outside the present diagnostic study.

G Mechanism-attribution probe

Reading. Descriptive mechanism check.

Section 5.6 reports what moves in the representation. To localize whether the perturbation enters through the encoder or later predictor/planner stages, we use latent-noise probing as a conservative mechanism check. Directly injecting noise into latent z skips the encoder and decouples encoder contributions from predictor and cost contributions.

Table 13: Three-layer attribution by task (LeWM $n = 9$ sweep, unified 3-seed $\times$ 100 eval). These probes are descriptive mechanism readouts, not checkpoint selectors.

Task	Strongest signal in these probes	Bounded interpretation
TwoRoom	encoder-shift signal (rank-saturated)	saturation limits mechanism inference
PushT	encoder + single-step predictor	contact-relevant resolution must be protected
Reacher	encoder + multi-step rollout	rollout stabilization aligns with the representative diagnostic movement
Cube	encoder	weaker behavioral recovery despite rollout stabilization

Across these probes, the strongest common signal is encoder-side perturbation transduced by the predictor. We use this as mechanism localization only: the sweep has broad plateaus and finite evaluation variance, so these scalar readouts should not be used to rank checkpoints inside a plateau. A stronger causal claim about planner or cost-surface contributions would require multi-task latent-injection or cost-ranking ablations.

H Cross-method replication: PLDM noise sweep on four tasks

This appendix records the second method-family on which the Gaussian-noise sweep and ACPC-basin diagnostics replicate. PLDM follows the joint-embedding predictive line from Sobal et al. [33]; the concrete latent-dynamics planning baseline is from Sobal et al. [34]. We use the implementation distributed through the `stable-worldmodel` baseline suite [25]. Training and evaluation follow the LeWM canonical protocol bit-for-bit: per-frame Gaussian pixel noise, $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$, three evaluation seeds (42/43/44), 100 trajectories per seed.

Coverage. The PLDM release is complete for the same grid as the LeWM canonical sweep: 4 tasks $\times$ 9 configurations (no-noise baseline plus $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$), three evaluation seeds (42/43/44), and 100 trajectories per seed. The PLDM eval, predictor-diagnostic, full-diagnostic, and ACPC-basin JSON files are listed in the data manifest; we do *not* merge them into the LeWM canonical files used by the main LeWM tables.

No-noise-trained PLDM cliff. PLDM reproduces the no-noise-trained Gaussian-noise cliff on PushT and TwoRoom, while Reacher and Cube show much smaller drops under this PLDM training recipe (Table 14). This supports a method-family reading of the failure mode without claiming every task has the same cliff magnitude.

Table 14: PLDM baseline trained without input-noise augmentation on all four tasks. Success rate mean $\pm$ population std, 3 evaluation seeds $\times$ 100 trajectories. Drop is clean success minus observation-noise 0.08 success.

Task	unpert.	obs 0.05	obs 0.08	obs+goal 0.08	drop
TwoRoom	96.67 ± 1.70	79.67 ± 4.03	67.00 ± 2.16	51.67 ± 2.05	29.67
PushT	75.33 ± 3.68	46.00 ± 4.97	18.33 ± 2.05	10.00 ± 2.16	57.00
Reacher	82.67 ± 1.70	81.33 ± 4.03	80.33 ± 5.73	79.33 ± 0.94	2.33
Cube	52.67 ± 3.30	50.33 ± 4.50	48.33 ± 4.50	48.33 ± 2.87	4.33

PLDM full sweep — unperturbed evaluation. Table 15 reports the per-task unperturbed success rate at every trained $\sigma_{\max}$ level. Table 16 reports the corresponding observation-noise 0.08 robustness with an unperturbed goal.

Table 15: PLDM+noise sweep, unperturbed success rate (%). † marks one representative high grid row per task for compact reading.

$\sigma_{\max}$	TwoRoom	PushT	Reacher	Cube
0 (base)	96.67 ± 1.70	75.33 ± 3.68	82.67 ± 1.70	52.67 ± 3.30
0.01	96.33 ± 0.94	72.33 ± 4.19	78.00 ± 0.82	51.33 ± 1.25
0.02	97.33 ± 0.47	71.33 ± 3.86	82.33 ± 2.36	53.33 ± 1.70
0.03	96.67 ± 1.70	$76.67 \pm 7.54^{\dagger}$	83.00 ± 3.74	52.67 ± 3.30
0.04	97.67 ± 0.47	68.67 ± 5.25	$85.00 \pm 2.16^{\dagger}$	54.00 ± 2.94
0.05	97.67 ± 1.25	74.33 ± 3.40	72.00 ± 1.41	54.00 ± 2.16
0.06	98.00 ± 0.82	70.33 ± 6.18	79.00 ± 0.82	$56.00 \pm 4.97^{\dagger}$
0.07	98.00 ± 0.82	66.67 ± 8.38	80.67 ± 2.62	53.67 ± 3.68
0.08	$98.33 \pm 0.94^{\dagger}$	68.00 ± 1.41	80.33 ± 1.25	54.00 ± 1.63

Table 16: PLDM+noise sweep, observation-noise 0.08 success rate with unperturbed goal (%). ‡ marks one representative high grid row per task for compact reading.

$\sigma_{\max}$	TwoRoom	PushT	Reacher	Cube
0 (base)	67.00 ± 2.16	18.33 ± 2.05	80.33 ± 5.73	48.33 ± 4.50
0.01	96.00 ± 1.63	23.33 ± 2.49	77.67 ± 3.40	51.33 ± 2.36
0.02	95.67 ± 0.94	47.00 ± 0.82	80.67 ± 5.25	51.00 ± 4.32
0.03	96.33 ± 1.25	$73.00 \pm 7.48^{\ddagger}$	$81.67 \pm 4.71^{\ddagger}$	54.33 ± 2.62
0.04	97.33 ± 0.94	66.67 ± 5.73	80.33 ± 2.87	$54.67 \pm 2.36^{\ddagger}$
0.05	97.67 ± 1.89	71.67 ± 5.79	62.67 ± 4.19	54.00 ± 3.74
0.06	$98.00 \pm 0.00^{\ddagger}$	69.33 ± 4.78	75.00 ± 0.82	54.33 ± 0.94
0.07	98.00 ± 0.82	64.00 ± 7.79	78.33 ± 2.05	53.00 ± 3.74
0.08	97.67 ± 1.25	68.00 ± 5.66	79.33 ± 1.25	53.67 ± 2.05

Reading. PLDM is a bounded second-family replication, not a competing baseline. It reproduces the large Gaussian cliff and recovery on TwoRoom and PushT: observation-noise 0.08 success rises from 67.00% to 98.00% on TwoRoom and from 18.33% to 73.00% on PushT. Reacher and Cube are low-cliff or weakly responsive under this PLDM recipe, matching the paper's task-dependent plateau reading. The exact high-grid location and clean/noisy tradeoff remain method-dependent, so PLDM strengthens the controlled fragility/recovery signature without making a method-superiority claim.

PLDM full-sweep ACPC basin replication. We also run the Gaussian-noise basin protocol on all 36 PLDM checkpoints. Table 17 reports the compact baseline-vs-representative observation-noise summary; the full 4 tasks $\times$ 9 configurations grid is released in the manifest-listed PLDM ACPC-basin artifact.

Table 17: PLDM full-sweep Gaussian-noise ACPC basin replication, summarized by the no-noise baseline and each task's representative high-noise PLDM observation-noise 0.08 grid row. The protocol matches Table 5: same-state Gaussian-noise views of the observation history (std 0.01–0.08), unperturbed goal, and the same recorded action sequence.

Task	checkpoint	obs 0.08	drop	R_E	R_F
TwoRoom	base	67.00	29.67	3.287	0.611
TwoRoom	noise 0.06	98.00	0.00	0.235	0.034
PushT	base	18.33	57.00	0.874	0.846
PushT	noise 0.03	73.00	3.67	0.153	0.091
Reacher	base	80.33	2.33	0.237	0.060
Reacher	noise 0.03	81.67	1.33	0.135	0.039
Cube	base	48.33	4.33	0.831	0.453
Cube	noise 0.04	54.67	−0.67	0.146	0.054

Basin reading. The PLDM basin replication is directionally consistent with the LeWM ACPC-basin reading: the representative observation-noise checkpoint reduces R_F on all four tasks, sharply on TwoRoom, PushT, and Cube. Reacher is the boundary case because PLDM is already near-robust at the no-noise baseline. This reduces the concern that the paired same-state perturbation pattern is purely a LeWM artifact, while leaving the mechanism claim bounded by the full-diagnostic table below.

PLDM full-diagnostic mechanism check. We also run the same five diagnostic layers on the PLDM sweep. The compact base-vs-representative view in Table 18 matches the guard reading in Table 8: rank, transition resolution, and inverse-dynamics probes remain broadly preserved at representative recovered rows, while rollout drift drops strongly. The important boundary is architecture dependence: LeWM compresses TwoRoom's effective rank where PLDM does not, so the shared claim is control-time fragility/recovery, not a universal internal mechanism.

Table 18: PLDM full-diagnostic check, no-noise baseline → representative noise-trained checkpoint. Representatives are high-noise PLDM observation-noise 0.08 grid rows (TwoRoom $\sigma_{\max} = 0.06$, PushT $\sigma_{\max} = 0.03$, Reacher $\sigma_{\max} = 0.03$, Cube $\sigma_{\max} = 0.04$).

Task	rep. $\sigma_{\max}$	eff. rank	trans. L2	ID probe R^2	8-step rollout L2
TwoRoom	0.06	74.78 → 72.70	0.8188 → 0.8254	0.3233 → 0.3201	20.36 → 1.02
PushT	0.03	130.13 → 131.90	0.4710 → 0.4778	0.7570 → 0.7479	20.25 → 2.82
Reacher	0.03	117.52 → 108.14	0.5053 → 0.4824	0.2150 → 0.1844	1.45 → 0.94
Cube	0.04	134.80 → 124.49	0.6395 → 0.6313	0.5428 → 0.5576	18.98 → 4.02

PLDM boundary. PLDM reduces the concern that the main pattern is tied to one LeWM checkpoint family. It also reinforces the bounded-plateau reading: representative high-grid locations are method- and task-dependent, and the diagnostic tables should localize robust regimes rather than rank points inside a broad sweep.

I Cross-perturbation sanity check: no-noise-trained blur evaluation

This appendix addresses the narrow concern that the observed visual fragility is an artefact of Gaussian noise only. We evaluate the LeWM and PLDM baselines trained without input-noise augmentation under Gaussian blur with kernel sizes $3, 7, 11, 15$, applied to pixels, goal images, or both. This is an **eval-only** stress test: no blur-trained checkpoint is included, and the blur data are not mixed into the Gaussian-noise sweep. The released source is listed in `DATA_MANIFEST.md`.

Table 19: Observation+goal blur stress test on baselines trained without input-noise augmentation. Success rate mean ± population std over 3 evaluation seeds × 100 trajectories. Kernel sizes 11 and 15 are severe low-pass endpoints on 224×224 inputs, so they should not be read as mild perturbations.

Method	Task	unpert.	$k = 3$	$k = 7$	$k = 11$	$k = 15$	worst drop
LeWM	TwoRoom	94.00 ± 3.56	39.33 ± 2.05	32.67 ± 3.09	30.33 ± 2.87	30.67 ± 0.47	63.67
LeWM	PushT	86.33 ± 2.36	81.67 ± 4.50	74.00 ± 1.41	65.33 ± 2.49	58.67 ± 6.65	27.67
LeWM	Reacher	58.67 ± 1.25	57.00 ± 6.38	45.67 ± 2.05	36.00 ± 4.90	20.33 ± 2.49	38.33
LeWM	Cube	66.67 ± 2.62	65.00 ± 1.63	62.33 ± 2.62	57.33 ± 2.87	54.00 ± 2.16	12.67
PLDM	TwoRoom	96.67 ± 1.70	38.33 ± 0.47	30.33 ± 2.87	28.33 ± 1.25	28.33 ± 1.70	68.33
PLDM	PushT	75.33 ± 3.68	74.00 ± 2.94	72.00 ± 3.56	67.67 ± 4.03	62.33 ± 2.05	13.00
PLDM	Reacher	82.67 ± 1.70	86.00 ± 2.16	80.00 ± 3.56	72.00 ± 2.16	68.00 ± 4.08	14.67
PLDM	Cube	52.67 ± 3.30	53.00 ± 5.35	53.00 ± 2.83	50.67 ± 3.68	52.33 ± 5.44	2.00

Reading. Blur is a different failure mode from Gaussian pixel noise. The clear collapse is TwoRoom: it falls below 40% already at $k = 3$ on both LeWM and PLDM and then saturates around 28–33% for $k = 7$–15. This reverses the Gaussian-noise ordering, where PushT is the most fragile task. The other tasks are more stable under blur: PLDM PushT, PLDM Reacher, and both Cube baselines lose at most about 15 pt at the strongest endpoint, while LeWM PushT degrades gradually and LeWM Reacher drops sharply only at $k = 15$. We therefore use blur only as an eval-only cross-perturbation sanity check: visual fragility is not unique to Gaussian noise, but the task ordering and recovery profile are perturbation-specific. Because this appendix does not train blur-robust checkpoints or test Gaussian-trained checkpoints under blur, it does not establish cross-perturbation transfer.

J Unseen-stressor score and matched diagnostic check

After the Gaussian development grid and training-seed replication were frozen, we ran strongest-severity unseen-stressor score checks on all three LeWM training seeds (3072/3073/3074), comparing no-noise training to $\sigma_{\max} = 0.08$ under two unseen visual stressors: Gaussian blur with kernel size 15 and resize/down-upsample with factor 0.25. A compact four-row summary of this appendix appears in Table 10; the appendix gives the full score aggregate and matched diagnostic slice. This appendix does not broaden the main Gaussian-noise claim into a cross-perturbation result. It asks whether the Gaussian-trained endpoint transfers to non-Gaussian stressors, and whether the paired diagnostics move in the same direction when the behavioral effect is clear. The score aggregate and selected diagnostics subset both include all three training seeds, with TwoRoom/Reacher blur as clear positive-transfer endpoints and PushT/Cube resize as small-effect endpoints. The full review file is listed in `DATA_MANIFEST.md`.

Table 20: Three-training-seed unseen-stressor score check. Values are mean ± population std across LeWM training seeds 3072/3073/3074; each seed point averages evaluation seeds 42/43/44 from the strongest-severity unseen runs (`num_eval=300`). Stress Δ is $\sigma_{\max} = 0.08$ stress success minus no-noise stress success; drop improvement is the reduction in clean-to-stress drop.

Task	unseen stress	baseline stress	$\sigma_{\max} = 0.08$ stress	stress Δ	drop impr.	reading
TwoRoom	blur $k = 15$	47.67 ± 5.44	90.78 ± 5.38	$+43.11 \pm 8.90$	$+40.89 \pm 8.23$	clear score gain
Reacher	blur $k = 15$	22.00 ± 3.78	71.22 ± 1.10	$+49.22 \pm 3.29$	$+30.22 \pm 4.16$	clear score gain
PushT	resize 0.25	63.44 ± 14.05	66.33 ± 8.38	$+2.89 \pm 17.98$	-3.78 ± 15.56	no clear gain
Cube	resize 0.25	57.00 ± 1.96	56.11 ± 0.57	-0.89 ± 1.40	$+2.78 \pm 3.40$	no clear gain

Reading. TwoRoom and Reacher show the desired agreement: the unseen stress scores improve strongly and the displayed paired diagnostics move in the better direction. PushT and Cube should not be read as negative transfer cases: their resize score deltas are small at the scale of the evaluation variance, so the behavioral conclusion is no clear effect. The diagnostics act as a specificity check for the same cautious interpretation. PushT shows improved ACPC-H/transition and PCC but almost unchanged CRA and no clear stress-score gain; Cube shows small and inconsistent ACPC/PCC/CRA movement. We therefore treat this as bounded appendix evidence for diagnostic localization on selected clear-effect unseen stressors, plus no coordinated score/diagnostic signal on the resize endpoints; it is not evidence that ACPC is a universal cross-perturbation robustness predictor.

K Phase-0 paired ACPC diagnostics

Reading. Exploratory artifact support.

The exploratory clean-goal paired ACPC release covers LeWM and PLDM, four tasks, and nine training-noise levels per task. It is released as `assets/paper1_data/acpc_phase0_clean_goal_seed9101.json`, with the three-seed LeWM extension in `assets/paper1_data/acpc_phase0_lewm_three_seed.json`. The PDF keeps the compact LeWM readout table in Table 6; the full ADM/SPRR/top-k release remains an exploratory artifact and is not used for model selection.

L Plateau-membership audit for the frozen diagnostic screen

This audit uses the same 96 nonzero LeWM rows as the three-training-seed Gaussian grid. It treats $\sigma_{\max}$ as a candidate label inside each task–training-seed block. The goal is not to rank checkpoints inside a plateau, but to test whether diagnostic scores screen a candidate region enriched for checkpoints inside the closed-loop robustness plateau. A checkpoint is a plateau member if its observation-noise 0.08 score is within 5pp of the block best. Each diagnostic screen returns the top half (4/8) of nonzero candidates in a block. The released artifact is `assets/paper1_data/selector_plateau_audit_20260704.json`; additional single-readout rows are reported in the released artifact.

Table 23: Plateau-membership audit. A checkpoint is a plateau member if its observation-noise 0.08 score is within 5pp of the block best. Each diagnostic screen returns four of eight nonzero candidates per task–training-seed block. Precision and recall are the primary readouts; presence is reported only to show block coverage.

Rule	precision	recall	TP/FP/FN	presence hit	reading
Aggregate ACPC/PCC/CRA/MAF	87.5%	61.8%	42/6/26	12/12	plateau-entry diagnostic screen
MAF only	91.7%	64.7%	44/4/24	12/12	competitive single-readout reference
High-std top-half reference	87.5%	61.8%	42/6/26	12/12	simple intervention-order reference
Random top-half reference	70.8%	50.0%	34.0/14.0/34.0	11.96/12	exact random reference

Reading. The audit evaluates membership enrichment, not point-optimal ranking. The aggregate screen improves precision over exact random top-half screening (87.5% vs. 70.8%), but does not exceed MAF-only or the high-std reference; this is not evidence of selector dominance. The high-std reference is a coarse intervention-order screen: larger noise often helps before plateau entry, but once a task–seed block reaches the plateau, $\sigma_{\max}$ is treated only as a candidate label, not as a monotone performance predictor. We therefore read the screen as diagnostic enrichment and failure localization.

M Reduced-budget offline CEM trace audit

The fixed-pool diagnostics deliberately share candidate sets, whereas the deployed planner updates a CEM proposal distribution. To partially close this planner gap without retraining or new closed-loop sweeps, we run the actual `stable_worldmodel.solver.CEMSolver` on logged states from the same base and fixed $\sigma_{\max} = 0.08$ checkpoints across training seeds 3072/3073/3074. This audit uses a reduced CEM budget (4 states per task–seed–endpoint, horizon 5, action block 5, 64 samples, 8 CEM iterations, top-8 elites) and the same solver seed for clean/noisy histories. It is an offline optimizer trace, not a replacement for closed-loop evaluation.

Table 24: Offline CEMSolver clean/noisy trace over training seeds 3072/3073/3074. Final plan L2/dim and first-action L2/dim compare the final clean/noisy CEM plans after the reduced-budget optimizer run. Seeded top-1 flip uses matched solver sample indices and is meaningful as a trace diagnostic, not as a full adaptive CEM guarantee.

Task	std	final plan L2/dim	first action L2/dim	final top-1 flip	final top-k Jaccard
TwoRoom	0.0	1.24 ± 0.14	0.98 ± 0.09	0.83 ± 0.12	0.16 ± 0.04
TwoRoom	0.08	0.61 ± 0.13	0.73 ± 0.24	0.92 ± 0.12	0.32 ± 0.09
PushT	0.0	1.96 ± 0.04	1.41 ± 0.13	1.00 ± 0.00	0.09 ± 0.01
PushT	0.08	0.64 ± 0.03	0.79 ± 0.13	0.75 ± 0.20	0.39 ± 0.10
Reacher	0.0	1.15 ± 0.04	1.17 ± 0.02	0.83 ± 0.24	0.22 ± 0.04
Reacher	0.08	0.38 ± 0.00	0.52 ± 0.05	0.25 ± 0.20	0.53 ± 0.06
Cube	0.0	1.30 ± 0.09	1.04 ± 0.01	1.00 ± 0.00	0.12 ± 0.04
Cube	0.08	0.51 ± 0.08	0.50 ± 0.03	0.42 ± 0.12	0.38 ± 0.12

Reading. The trace supports the same direction as the fixed-pool diagnostics for plan displacement and best-cost stability: the fixed high-noise endpoint reduces final clean/noisy plan displacement and final best-cost deltas on every task. It does not prove complete adaptive CEM stability: TwoRoom still has high final seeded top-1 flip despite lower plan displacement and higher top-k overlap. We therefore use this audit only to reduce the fixed-pool/planner gap, not to replace closed-loop evaluation or the fixed-candidate theory limits.

N Target-view ablation completion check

This appendix records the completed LeWM target-view ablation referenced in Section 6. The runs use the origin-future target configuration: the prediction context is encoded from the configured Gaussian-perturbed history, while the prediction target remains the original future latent. This is the perturbed-history → original-future branch, contrasted with the default full-sequence perturbed-target branch used by the main LeWM+noise sweep. The released summary artifact is listed in `DATA_MANIFEST.md`.

We verified the four-task run set after completion: eight target-noise checkpoints per task and epoch-10 summaries for unperturbed, observation-noise 0.03, observation-noise 0.05, and observation-noise 0.08 evaluation. The canonical table below uses seeds 42/43/44 and 100 trajectories per seed.

Table 25: LeWM perturbed-history → original-future target-view ablation. Values are success rate mean ± population std over the eight target-noise checkpoints; each checkpoint value is the three-seed mean from `eval_summary.csv`.

Task	unperturbed	obs 0.03	obs 0.05	obs 0.08
TwoRoom	93.33 ± 1.72	88.67 ± 6.04	77.71 ± 9.12	61.75 ± 7.65
PushT	82.92 ± 4.00	57.71 ± 8.96	20.21 ± 11.68	6.75 ± 5.00
Reacher	59.92 ± 1.80	40.38 ± 2.78	32.00 ± 4.57	19.62 ± 3.97
Cube	65.00 ± 2.01	61.42 ± 2.06	44.29 ± 1.32	39.62 ± 1.22

Table 26: Closed-loop target-view comparison at observation-noise 0.08. Values are success rate mean ± population std over the eight training-noise checkpoints; each checkpoint value is the canonical three-seed mean.

Task	full-sequence perturbed target	perturbed-history → original future	full-seq advantage
TwoRoom	94.71 ± 1.90	61.75 ± 7.65	+32.96
PushT	72.83 ± 12.84	6.75 ± 5.00	+66.08
Reacher	76.17 ± 10.89	19.62 ± 3.97	+56.54
Cube	59.83 ± 5.55	39.62 ± 1.22	+20.21

Reading. The ablation acts as a negative scope check and supports retaining the full-sequence empirical branch. Simply changing the prediction target to the original future does not reproduce the recovery from input-side noise augmentation in Table 2 and fig. 2: PushT and Reacher remain close to their no-noise robustness cliffs, TwoRoom degrades under stronger eval noise despite high unperturbed success, and Cube does not improve over the main sweep.

A matched PushT 0to008 check shows the contrast: rerunning both branches' $\sigma_{\max} = 0.08$ checkpoints under the same fixed evaluation code and canonical seeds gives observation-noise 0.08 three-seed means of 86.33% for the full-sequence checkpoint versus 8.67% for the origin-target checkpoint; the canonical sweep artifact gives 89.00% for the same full-sequence checkpoint, and the difference is normal evaluation variance. We interpret this as evidence consistent with a closed-loop train/eval consistency explanation: full-sequence noise augmentation better matches the planner's latent autoregressive feedback, while one-step origin-target denoising does not. The result narrows the mechanism: target view alone is not the missing ingredient, and an explicit action-conditioned predictive-consistency objective is the method-paper path suggested by the diagnostics.

Table 21: Three-seed unseen-stressor matched diagnostic slice with selected Phase-0 paired diagnostics. Values are averages over training seeds 3072/3073/3074. Stress delta is $\sigma_{\max} = 0.08$ minus the no-noise baseline at the strongest unseen stress. Drop improvement is the reduction in clean-to-stress drop. Negative Δ is better for ACPC-H/transition and PCC; positive Δ is better for CRA. Small PushT/Cube score deltas are treated as no clear behavioral effect at this evaluation scale.

Task	stress	stress Δ	drop impr.	Δ ACPC	Δ PCC	Δ CRA	reading
TwoRoom	blur $k = 15$	+43.11	+40.89	−0.691	−44.7	+0.616	clear signal
Reacher	blur $k = 15$	+49.22	+30.22	−1.895	−48.9	+0.587	clear signal
PushT	resize 0.25	+2.89	−3.78	−0.267	−6.7	+0.010	no clear effect
Cube	resize 0.25	−0.89	+2.78	+0.052	−0.6	−0.030	no clear effect

Table 22: Full blur/resize unseen-stressor diagnostic audit. Rows are the 24 task–family–training-seed endpoints: four tasks, blur and resize strongest stressors, and training seeds 3072/3073/3074. Correlations compare diagnostic deltas from $\sigma_{\max} = 0.0 \rightarrow 0.08$ against closed-loop strongest-stress score deltas and drop improvements. This table reduces selected-slice risk but remains a bounded two-family scope check.

Metric	ρ vs stress Δ	r vs stress Δ	ρ vs drop impr.	r vs drop impr.	n
ACPC-H/trans. Δ	0.92	0.85	0.77	0.67	24
PCC Δ	0.94	0.90	0.84	0.90	24
CRA Δ	0.94	0.89	0.87	0.91	24
MAF Δ	0.90	0.91	0.77	0.77	24
Composite signed-rank	0.94	0.94	0.82	0.84	24